MATH 419: Stochastic Processes

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Chapter 1

Conditional Expectation

Definition. A measure μ on a σ-field $(\Omega, \mathcal{F})$ is **σ-finite** if there exists a sequence $(A_n)_{n\in\mathbb{N}}$ such that $A_n \in \mathcal{F}$, $A_n \subset A_{n+1}$, $\mu(A_n) < \infty$ for all $n \in \mathbb{N}$, and $\bigcup_{n\in\mathbb{N}} A_n = \Omega$ (abbreviated $A_n \uparrow \Omega$).
If μ, ν are measures on $(\Omega, \mathcal{F})$, we say ν is **absolutely continuous** with respect to μ, written $\nu \ll \mu$, if for all $A \in \mathcal{F}$, $\mu(A) = 0$ implies $\nu(A) = 0$.

Example. Let $X \geq 0$ be a random variable on a probability space $(\Omega, \mathcal{F}, P)$. Define

$$\nu(A) = \int_A X \,\mathrm{d}P = \int_\Omega \mathbf{1}_A X \,\mathrm{d}P, \quad A \in \mathcal{F}.$$

Then ν is a σ-finite measure on $(\Omega, \mathcal{F})$ with $\nu \ll P$. Indeed, setting $A_n = \{X \leq n\} \uparrow \Omega$ gives $\nu(A_n) = \int_{A_n} X \,\mathrm{d}P \leq n < \infty$; and $P(A) = 0$ implies $X\mathbf{1}_A = 0$ P-a.s., so $\nu(A) = 0$.

Theorem (Radon–Nikodym theorem). Let ν, μ be two non-negative, σ-finite measures on $(\Omega, \mathcal{F})$ with $\nu \ll \mu$. Then:

(i) There exists a random variable $X : \Omega \to [0, \infty)$ such that $\nu(A) = \int_A X \,\mathrm{d}\mu$ for all $A \in \mathcal{F}$.

(ii) If X' is another such random variable satisfying (i), then $X = X'$ μ-almost surely.

We write $X = \dfrac{\mathrm{d}\nu}{\mathrm{d}\mu}$ and call it the **Radon–Nikodym derivative**.

Proof. (i) This is part of measure theory (MATH 420). See Appendix A.4 in Durrett.
(ii) Let (B_n) be an increasing sequence of measurable sets with $\mu(B_n) < \infty$ and $B_n \uparrow \Omega$ (from σ-finiteness of μ). For each n, let

$$A_n = B_n \cap \left\{ n \geq X \geq X' + \tfrac{1}{n} \right\}.$$

The σ-finiteness of ν implies $X < \infty$ and $X' < \infty$ μ-a.e., so the expressions below are well-defined.[1] Then

$$\frac{1}{n}\mu(A_n) \leq \int_{A_n} (X - X')\,\mathrm{d}\mu = \int X\mathbf{1}_{A_n}\,\mathrm{d}\mu - \int X'\mathbf{1}_{A_n}\,\mathrm{d}\mu = \nu(A_n) - \nu(A_n) = 0.$$

Hence $\mu(A_n) = 0$ for all n. Therefore $\mu(\{X > X'\}) = \mu\left(\bigcup_{n\in\mathbb{N}} A_n\right) = 0$, i.e. $X \leq X'$ μ-a.s. By symmetry, $X \geq X'$ μ-a.s., so $X = X'$ μ-a.s. □

Exercise. (i) If $\nu \ll \mu$ are non-negative σ-finite measures on $(\Omega, \mathcal{F})$ with $\dfrac{\mathrm{d}\nu}{\mathrm{d}\mu} = f$, show that g is ν-integrable (i.e. $\int |g|\,\mathrm{d}\nu < \infty$) if and only if gf is μ-integrable, and in that case $\int g\,\mathrm{d}\nu = \int gf\,\mathrm{d}\mu$.

(ii) If μ, λ, ν are non-negative σ-finite measures with $\lambda \ll \mu$ and $\nu \ll \lambda$, show that $\nu \ll \mu$ and $\dfrac{\mathrm{d}\nu}{\mathrm{d}\mu} = \dfrac{\mathrm{d}\nu}{\mathrm{d}\lambda}\dfrac{\mathrm{d}\lambda}{\mathrm{d}\mu}$ μ-a.e.

Notation. We write $\mathcal{D} \overset{\sigma}{\subset} \mathcal{F}$ to mean $\mathcal{D}$ is a sub-σ-field of $\mathcal{F}$.

Theorem (Existence and uniqueness of conditional expectation). Let $\mathcal{D} \overset{\sigma}{\subset} \mathcal{F}$ be σ-fields over Ω, and let X be an integrable random variable on $(\Omega, \mathcal{F}, P)$. There exists a unique (up to P-null sets) random variable $E[X|\mathcal{D}]$ satisfying:

(i) $E[X|\mathcal{D}]$ is $\mathcal{D}$-measurable.

(ii) $\displaystyle\int_D E[X|\mathcal{D}]\,\mathrm{d}P = \int_D X\,\mathrm{d}P$ for all $D \in \mathcal{D}$.

Proof. **Existence.** First assume $X \geq 0$. Define

$$\nu(D) = \int_D X\,\mathrm{d}P, \quad D \in \mathcal{D}.$$

Then $\nu \ll P$ on $(\Omega, \mathcal{D})$, and ν is σ-finite. By the Radon–Nikodym theorem (Theorem 1), there exists $E[X|\mathcal{D}] := \dfrac{\mathrm{d}\nu}{\mathrm{d}P}$ on $(\Omega, \mathcal{D})$ satisfying (i) and (ii). For general X: write $X = X^+ - X^-$ where $X^+ = \max(X, 0)$ and $X^- = \max(-X, 0)$. Define

$$E[X|\mathcal{D}] := E[X^+|\mathcal{D}] - E[X^-|\mathcal{D}],$$

and verify (i) (linearity of measurability) and (ii) (linearity of integrals).

[1] The condition $X \leq n$ ensures $\int X\mathbf{1}_{A_n}\mathrm{d}\mu$ and $\int X'\mathbf{1}_{A_n}\mathrm{d}\mu$ are finite, avoiding the ill-defined expression $\nu(A_n) - \nu(A_n) = \infty - \infty$.

Uniqueness. Let $E[X|\mathcal{D}]$ and $\widetilde{E}[X|\mathcal{D}]$ both satisfy (i) and (ii). For each $n \in \mathbb{N}$, let

$$D_n = \left\{\omega \in \Omega : E[X|\mathcal{D}](\omega) - \widetilde{E}[X|\mathcal{D}](\omega) \geq \tfrac{1}{n}\right\} \in \mathcal{D}.$$

Then

$$P(D_n) \leq n \int_{D_n} \Big(E[X|\mathcal{D}] - \widetilde{E}[X|\mathcal{D}]\Big)\,\mathrm{d}P = n\left(\int_{D_n} X\,\mathrm{d}P - \int_{D_n} X\,\mathrm{d}P\right) = 0.$$

Since $\bigcup_{n\in\mathbb{N}} D_n = \{E[X|\mathcal{D}] > \widetilde{E}[X|\mathcal{D}]\}$, we have $E[X|\mathcal{D}] \leq \widetilde{E}[X|\mathcal{D}]$ P-a.s. By symmetry, equality holds a.s. $\square$

Definition (Conditional expectation). Let $(\Omega, \mathcal{F}, P)$ be a probability space and $\mathcal{D} \overset{\sigma}{\subset} \mathcal{F}$. Given an integrable random variable X, the random variable $E[X|\mathcal{D}]$ from Theorem 1 is called the **conditional expectation of X given $\mathcal{D}$**. If Y is a random variable, we write $E[X|Y] := E[X|\sigma(Y)]$.

Remark. (1) $E[X|\mathcal{D}]$ is unique only up to P-null sets.

(2) If $X \geq 0$ is a (not necessarily integrable) random variable and $\mathcal{D} \overset{\sigma}{\subset} \mathcal{F}$, then $\nu(D) = \int_D X\,\mathrm{d}P$ is σ-finite with $\nu \ll P$ on $(\Omega, \mathcal{D})$. The Radon–Nikodym derivative $\dfrac{\mathrm{d}\nu}{\mathrm{d}P}$, defined up to P-null sets, serves as $E[X|\mathcal{D}]$.

(3) Integrability of X implies integrability of $E[X|\mathcal{D}]$. Indeed, using the non-negative case:

$$\int_\Omega |E[X|\mathcal{D}]|\,\mathrm{d}P \leq \int_\Omega E[X^+|\mathcal{D}]\,\mathrm{d}P + \int_\Omega E[X^-|\mathcal{D}]\,\mathrm{d}P = \int_\Omega X^+\,\mathrm{d}P + \int_\Omega X^-\,\mathrm{d}P = \int_\Omega |X|\,\mathrm{d}P < \infty.$$

(This also follows from conditional Jensen's inequality, Corollary 1.)

Example (Partition). Let $(\Omega, \mathcal{F}, P)$ be a probability space and $\Omega = \bigsqcup_{i=1}^{n} B_i$ a measurable partition with $B_i \cap B_j = \emptyset$ for $i \neq j$, $B_i \in \mathcal{F}$, $P(B_i) > 0$. Let $\mathcal{D} = \sigma(\{B_1, \ldots, B_n\})$ and define

$$E[X|B_i] := \frac{1}{P(B_i)} \int_{B_i} X\,\mathrm{d}P.$$

Then $E[X|\mathcal{D}] = \sum_{i=1}^{n} E[X|B_i]\,\mathbf{1}_{B_i}(\omega)$.

Proof. The function $Y(\omega) := \sum_{i=1}^{n} E[X|B_i]\,\mathbf{1}_{B_i}(\omega)$ is $\mathcal{D}$-measurable. For any non-empty $D \in \mathcal{D}$, write $D = \bigsqcup_{k=1}^{m} B_{i_k}$. Then

$$\int_D Y\,\mathrm{d}P = \sum_{k=1}^{m} E[X|B_{i_k}]\,P(B_{i_k}) = \sum_{k=1}^{m} \int_{B_{i_k}} X\,\mathrm{d}P = \int_D X\,\mathrm{d}P. \qquad \square$$

In particular, taking $X = \mathbf{1}_A$ gives $E[\mathbf{1}_A|\mathcal{D}](\omega) = \sum_{i=1}^{n} P(A|B_i)\,\mathbf{1}_{B_i}(\omega)$ P-a.s., which recovers the classical conditional probability.

Example. If X is $\mathcal{D}$-measurable and integrable, then $E[X|\mathcal{D}] = X$ P-a.s. (Both sides satisfy (i) and (ii) of Definition 1.)

Definition (Independence). Let $(\Omega, \mathcal{F}, P)$ be a probability space.

- Events $A, B \in \mathcal{F}$ are **independent** if $P(A \cap B) = P(A)P(B)$.
- Sub-σ-fields $\mathcal{G}, \mathcal{H} \overset{\sigma}{\subset} \mathcal{F}$ are **independent** if for all $A \in \mathcal{G}$, $B \in \mathcal{H}$: $P(A \cap B) = P(A)P(B)$.
- A random variable X and a sub-σ-field $\mathcal{G}$ are independent if $\sigma(X)$ and $\mathcal{G}$ are independent.
- Random variables X, Y are independent if $\sigma(X)$ and $\sigma(Y)$ are independent.

Example (Conditioning on independent σ-field). Let X be integrable and independent of $\mathcal{D}$. Then $E[X|\mathcal{D}] = E[X]$ P-a.s.

Proof. For all $D \in \mathcal{D}$:

$$\int_D X \,\mathrm{d}P = \int \mathbf{1}_D X \,\mathrm{d}P = \int \mathbf{1}_D \,\mathrm{d}P \cdot \int X \,\mathrm{d}P = P(D)\,E[X] = \int_D E[X] \,\mathrm{d}P,$$

where the second equality uses independence of X and $\mathbf{1}_D$ (since $\mathbf{1}_D$ is $\mathcal{D}$-measurable). The constant $E[X]$ is $\mathcal{D}$-measurable, so $E[X|\mathcal{D}] = E[X]$ a.s. $\square$

Intuition: $E[X|\mathcal{D}]$ is the best guess of X given the information $\mathcal{D}$. If $\mathcal{D}$ tells us nothing about X, the best guess is the mean.

An important special case of Example 1 is the trivial σ-field $\mathcal{D} = \{\emptyset, \Omega\}$. Every random variable is independent of $\mathcal{D}$, and every $\mathcal{D}$-measurable random variable is a.s. constant. Hence $E[X|\mathcal{D}] = E[X]$ a.s., showing that conditional expectation generalises expectation.

Proposition (Elementary properties). Let $(\Omega, \mathcal{F}, P)$ be a probability space, X_1, X_2 integrable, $\mathcal{D} \overset{\sigma}{\subset} \mathcal{E} \overset{\sigma}{\subset} \mathcal{F}$, and $a, b \in \mathbb{R}$.

(a) **Linearity:** $E[aX_1 + bX_2|\mathcal{D}] = aE[X_1|\mathcal{D}] + bE[X_2|\mathcal{D}]$ a.s.

(b) **Monotonicity:** $X_1 \leq X_2$ a.s. implies $E[X_1|\mathcal{D}] \leq E[X_2|\mathcal{D}]$ a.s.

(c) **Tower property / iterated conditioning:** $E[E[X_1|\mathcal{E}]|\mathcal{D}] = E[X_1|\mathcal{D}]$ a.s.

Proof. (a) For any $D \in \mathcal{D}$:

$$\int_D (aE[X_1|\mathcal{D}] + bE[X_2|\mathcal{D}])\,\mathrm{d}P = a\int_D X_1\,\mathrm{d}P + b\int_D X_2\,\mathrm{d}P = \int_D (aX_1 + bX_2)\,\mathrm{d}P.$$

Since $aE[X_1|\mathcal{D}] + bE[X_2|\mathcal{D}]$ is $\mathcal{D}$-measurable, we conclude by uniqueness.

(b) By (a), $E[X_2|\mathcal{D}] - E[X_1|\mathcal{D}] = E[X_2 - X_1|\mathcal{D}]$ a.s. Let $D_n = \{E[X_2 - X_1|\mathcal{D}] \le -\frac{1}{n}\} \in \mathcal{D}$. Then

$$-\frac{1}{n}P(D_n) \ge \int_{D_n} E[X_2 - X_1|\mathcal{D}]\,\mathrm{d}P = \int_{D_n} (X_2 - X_1)\,\mathrm{d}P \ge 0,$$

so $P(D_n) = 0$ for all n. Therefore $P(\{E[X_2|\mathcal{D}] < E[X_1|\mathcal{D}]\}) = P\left(\bigcup_n D_n\right) = 0$.

(c) Note $E[E[X_1|\mathcal{E}]|\mathcal{D}]$ is well-defined by Remark (3). For any $D \in \mathcal{D} \subset \mathcal{E}$:

$$\int_D E[E[X_1|\mathcal{E}]|\mathcal{D}]\,\mathrm{d}P = \int_D E[X_1|\mathcal{E}]\,\mathrm{d}P = \int_D X_1\,\mathrm{d}P.$$

Since $E[E[X_1|\mathcal{E}]|\mathcal{D}]$ is $\mathcal{D}$-measurable, we have $E[E[X_1|\mathcal{E}]|\mathcal{D}] = E[X_1|\mathcal{D}]$ a.s. □

Note: Proposition 1(b) immediately gives: $X_1 = X_2$ a.s. implies $E[X_1|\mathcal{D}] = E[X_2|\mathcal{D}]$ a.s.

Corollary (Conditional Jensen's inequality). Let $\varphi : \mathbb{R} \to \mathbb{R}$ be convex and X integrable with $\varphi(X)$ also integrable, and let $\mathcal{D} \overset{\sigma}{\subset} \mathcal{F}$. Then

$$E[\varphi(X)|\mathcal{D}] \ge \varphi(E[X|\mathcal{D}]) \quad P\text{-a.s.}$$

Proof. Every convex $\varphi : \mathbb{R} \to \mathbb{R}$ is the supremum of a countable collection of affine functions: there exists countable $S \subset \mathbb{R}^2$ such that $\varphi(x) = \sup_{(a,b)\in S} (ax + b)$ for all $x \in \mathbb{R}$.

For each $(a, b) \in S$, since $\varphi(x) \ge ax + b$ for all x, Proposition 1(a),(b) give

$$E[\varphi(X)|\mathcal{D}] \ge E[aX + b|\mathcal{D}] = aE[X|\mathcal{D}] + b \quad P\text{-a.s.}$$

Since S is countable, there is a single P-null set off which the above holds for all $(a, b) \in S$ simultaneously. Taking the supremum over all $(a, b) \in S$:

$$E[\varphi(X)|\mathcal{D}] \ge \sup_{(a,b)\in S} (aE[X|\mathcal{D}] + b) = \varphi(E[X|\mathcal{D}]) \quad P\text{-a.s.}$$ □

Exercise. Let $\mathcal{D} \overset{\sigma}{\subset} \mathcal{F}$ and X be square integrable ($E[X^2] < \infty$).

(i) Show $E[X|\mathcal{D}]$ is also square integrable. (Hint: conditional Jensen's inequality with $\varphi(x) = x^2$.)

(ii) For any square integrable $\mathcal{D}$-measurable Y, show

$$E[(X - Y)^2] \ge E[(X - E[X|\mathcal{D}])^2],$$

with equality iff $Y = E[X|\mathcal{D}]$ a.s. (Hint: write $X - Y = (X - E[X|\mathcal{D}]) + (E[X|\mathcal{D}] - Y)$ and expand the square.)

This exercise shows that $E[X|\mathcal{D}]$ is the *best* L^2 approximation to X among all $\mathcal{D}$-measurable random variables (orthogonal projection).

Theorem (Convergence theorems for conditional expectations). Let $(\Omega, \mathcal{F}, P)$ be a probability space and $\mathcal{D} \overset{\sigma}{\subset} \mathcal{F}$.

(a) **Conditional MCT:** If $0 \leq X_n \uparrow X$ and $E[X] < \infty$, then $E[X_n|\mathcal{D}] \uparrow E[X|\mathcal{D}]$ a.s.

(b) **Conditional Fatou:** If $X_n \geq 0$, $E[X_n] < \infty$, and $E[\liminf_n X_n] < \infty$, then

$$E\left[\liminf_{n\to\infty} X_n \,\middle|\, \mathcal{D}\right] \leq \liminf_{n\to\infty} E[X_n|\mathcal{D}] \quad \text{a.s.}$$

(c) **Conditional DCT:** If $X_n \to X$ a.s., $|X_n| \leq Y$ for all n with $E[Y] < \infty$, then

$$\lim_{n\to\infty} E[X_n|\mathcal{D}] = E[X|\mathcal{D}] \quad \text{a.s.}$$

Proof. (a) By monotonicity (Prop. 1(b)), the sequence $E[X_n|\mathcal{D}]$ is non-decreasing a.s.; let $U = \lim_{n\to\infty} E[X_n|\mathcal{D}]$ (defined a.s.). Since $E[X_n|\mathcal{D}] \leq E[X|\mathcal{D}]$ a.s. for each n, we have $U \leq E[X|\mathcal{D}]$ a.s. For any $D \in \mathcal{D}$, by the (ordinary) MCT:

$$\int_D U \,\mathrm{d}P = \lim_{n\to\infty} \int_D E[X_n|\mathcal{D}] \,\mathrm{d}P = \lim_{n\to\infty} \int_D X_n \,\mathrm{d}P = \int_D X \,\mathrm{d}P.$$

Since U is $\mathcal{D}$-measurable, $U = E[X|\mathcal{D}]$ a.s.
(b) Recall $\inf_{k\geq N} X_k \uparrow \liminf X_n$ as $N \to \infty$. For any $n \geq N$, $E[X_n|\mathcal{D}] \geq E[\inf_{k\geq N} X_k|\mathcal{D}]$ a.s. by monotonicity, so

$$\inf_{n\geq N} E[X_n|\mathcal{D}] \geq E\left[\inf_{n\geq N} X_n \,\middle|\, \mathcal{D}\right] \quad \text{a.s.}$$

Let $N \to \infty$ and apply part (a) to the sequence $\inf_{n\geq N} X_n \uparrow \liminf X_n$:

$$\liminf_{n\to\infty} E[X_n|\mathcal{D}] = \lim_{N\to\infty} \inf_{n\geq N} E[X_n|\mathcal{D}] \geq \lim_{N\to\infty} E\left[\inf_{n\geq N} X_n \,\middle|\, \mathcal{D}\right] = E\left[\liminf_n X_n \,\middle|\, \mathcal{D}\right].$$

(c) This follows from (b) by the same argument as the ordinary DCT. Indeed, a.s.

$$Y - X = \liminf_{n\to\infty}(Y - X_n) \quad \text{and} \quad Y - X_n \geq 0.$$

Applying (b):

$$E[Y - X|\mathcal{D}] \leq \liminf_{n\to\infty} E[Y - X_n|\mathcal{D}] = E[Y|\mathcal{D}] - \limsup_{n\to\infty} E[X_n|\mathcal{D}].$$

By linearity, $E[X|\mathcal{D}] \geq \limsup_n E[X_n|\mathcal{D}]$ a.s. Replacing X_n, X by $-X_n, -X$ and repeating gives $E[X|\mathcal{D}] \leq \liminf_n E[X_n|\mathcal{D}]$ a.s. Hence $E[X_n|\mathcal{D}] \to E[X|\mathcal{D}]$ a.s. $\square$

Proposition (Pull-out / measurability property). Let $(\Omega, \mathcal{F}, P)$ be a probability space, $\mathcal{D} \overset{\sigma}{\subset} \mathcal{F}$. Let Z be $\mathcal{D}$-measurable and X be $\mathcal{F}$-measurable such that X and XZ are both integrable. Then

$$E[XZ|\mathcal{D}] = Z\,E[X|\mathcal{D}] \quad \text{a.s.}$$

Proof. Without loss of generality we may assume $X, Z \geq 0$ (the general case follows by writing $X = X^+ - X^-$, $Z = Z^+ - Z^-$ and applying linearity). Since $ZE[X|\mathcal{D}]$ is $\mathcal{D}$-measurable (product of two $\mathcal{D}$-measurable functions), it suffices to verify that $\displaystyle\int_D XZ\,\mathrm{d}P = \int_D ZE[X|\mathcal{D}]\,\mathrm{d}P$ for all $D \in \mathcal{D}$.

Case 1: $Z = \mathbf{1}_{D'}$ for some $D' \in \mathcal{D}$. Then

$$\int_D XZ\,\mathrm{d}P = \int_{D\cap D'} X\,\mathrm{d}P = \int_{D\cap D'} E[X|\mathcal{D}]\,\mathrm{d}P = \int_D Z\,E[X|\mathcal{D}]\,\mathrm{d}P,$$

using that $D \cap D' \in \mathcal{D}$.

Case 2: $Z = \displaystyle\sum_{i=1}^{n} \alpha_i \mathbf{1}_{D_i}$, a non-negative simple function with $D_i \in \mathcal{D}$, $\alpha_i \geq 0$. This follows from Case 1 and linearity of integrals.

Case 3: General $Z \geq 0$. Take simple functions $Z_n \uparrow Z$. By Case 2, $\displaystyle\int_D XZ_n\,\mathrm{d}P = \int_D Z_n E[X|\mathcal{D}]\,\mathrm{d}P$. Since $E[X|\mathcal{D}] \geq 0$ a.s. (by Prop. 1(b)), let $n \to \infty$ and apply the ordinary MCT on both sides. □

Lemma (Doob–Dynkin Lemma). Let $Y : (\Omega, \mathcal{F}) \to (S, \mathcal{S})$ and $Z : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B})$ be measurable. Then Z is $\sigma(Y)$-measurable if and only if $Z = \varphi(Y)$ for some measurable $\varphi : (S, \mathcal{S}) \to (\mathbb{R}, \mathcal{B})$.

Proof. ($\Leftarrow$) $\varphi(Y)$ is the composition of measurable functions, hence $\sigma(Y)$-measurable.
($\Rightarrow$) Assume Z is $\sigma(Y)$-measurable. We proceed by the standard approximation argument.

Case 1: Z simple. $Z = \displaystyle\sum_{i=1}^{n} \alpha_i \mathbf{1}_{D_i}$ with $D_i \in \sigma(Y)$. Since $D_i \in \sigma(Y)$, there exists $S_i \in \mathcal{S}$ with $D_i = Y^{-1}(S_i)$. Then $Z = \varphi(Y)$ where $\varphi = \displaystyle\sum_{i=1}^{n} \alpha_i \mathbf{1}_{S_i} : (S, \mathcal{S}) \to (\mathbb{R}, \mathcal{B})$.

Case 2: $Z \geq 0$. Let $Z_n \uparrow Z$ be non-decreasing simple functions. By Case 1, $Z_n = \varphi_n(Y)$. Set $\varphi = \limsup_{n\to\infty} \max(\varphi_1, \ldots, \varphi_n)$. Then $\varphi \geq 0$ is measurable and $Z = \varphi(Y)$.

Case 3: General Z. Write $Z = Z^+ - Z^-$ and apply Case 2: $Z^+ = \varphi^+(Y)$, $Z^- = \varphi^-(Y)$, so $Z = (\varphi^+ - \varphi^-)(Y)$. □

The Doob–Dynkin lemma motivates the following definition.

Definition. Let $Y : (\Omega, \mathcal{F}) \to (S, \mathcal{S})$ be measurable and $Z : (\Omega, \mathcal{F}) \to \mathbb{R}$ integrable. By the Doob–Dynkin lemma, there exists a measurable $h : (S, \mathcal{S}) \to \mathbb{R}$ (unique up to P_Y-null sets, where P_Y is the law of Y) such that $E[Z|Y](\omega) = h(Y(\omega))$ P-a.s. We denote $h(y) = E[Z|Y = y]$.

Note: typically $P(Y = y) = 0$, so $E[Z|Y = y]$ cannot be interpreted as $\dfrac{E[Z\mathbf{1}_{\{Y=y\}}]}{P(Y = y)}$, but it is well-defined as the function h above.

Example (Continuous joint distribution). Let X, Y be real-valued random variables with joint density $f(x, y)$. Define the marginal density $f_Y(y) = \displaystyle\int f(x, y)\,\mathrm{d}x$ and the conditional density

$$f_{X|Y}(x|y) = \begin{cases} f(x, y)/f_Y(y) & \text{if } f_Y(y) > 0, \\ 0 & \text{if } f_Y(y) = 0. \end{cases}$$

Note $\displaystyle\int f_{X|Y}(x|y)\,\mathrm{d}x = 1$ when $f_Y(y) > 0$. For any Borel g with $E[|g(X)|] < \infty$:

$$E[g(X)|Y = y] = \int f_{X|Y}(x|y)\, g(x)\,\mathrm{d}x. \tag{1.5}$$

Proof. Without loss of generality $g \geq 0$. Define $h(y) = \displaystyle\int f_{X|Y}(x|y)\, g(x)\,\mathrm{d}x$ (Borel by Fubini). We verify $h(Y) = E[g(X)|Y]$ a.s., i.e. $\displaystyle\int \mathbf{1}_B(Y)\, h(Y)\,\mathrm{d}P = \int \mathbf{1}_B(Y)\, g(X)\,\mathrm{d}P$ for all $B \in \mathcal{B}(\mathbb{R})$:

$$\begin{aligned} \int \mathbf{1}_B(Y)\, h(Y)\,\mathrm{d}P &= \int \mathbf{1}_B(y)\, h(y)\, f_Y(y)\,\mathrm{d}y = \iint \mathbf{1}_B(y)\, f_{X|Y}(x|y)\, g(x)\, f_Y(y)\,\mathrm{d}x\,\mathrm{d}y \\ &= \iint \mathbf{1}_B(y)\, g(x)\, f(x, y)\,\mathrm{d}x\,\mathrm{d}y = \int \mathbf{1}_B(Y)\, g(X)\,\mathrm{d}P. \end{aligned}$$

□

Chapter 2

Martingales

Throughout this section, $(\Omega, \mathcal{F}, P)$ denotes the underlying probability space.

Definition. Let $(T, \leq)$ be a partially ordered set (typically $T = \mathbb{Z}_{\geq 0} = \{0, 1, 2, \ldots\}$ or $T = [0, \infty)$). A **filtration** $(\mathcal{F}_t)_{t \in T}$ is an increasing collection of sub-σ-fields of $\mathcal{F}$, i.e. $t_1 \leq t_2 \Rightarrow \mathcal{F}_{t_1} \overset{\sigma}{\subset} \mathcal{F}_{t_2}$.
A stochastic process $X = (X_t)_{t \in T}$ is a collection of random variables indexed by T. The process X is $(\mathcal{F}_t)$**-adapted** if X_t is $\mathcal{F}_t$-measurable for all $t \in T$.
An $(\mathcal{F}_t)$-adapted process X is an:

- $(\mathcal{F}_t)$**-supermartingale** if $E[|X_t|] < \infty$ for all t and $E[X_t|\mathcal{F}_s] \leq X_s$ a.s. for all $s \leq t$.
- $(\mathcal{F}_t)$**-submartingale** if $E[|X_t|] < \infty$ for all t and $E[X_t|\mathcal{F}_s] \geq X_s$ a.s. for all $s \leq t$.
- $(\mathcal{F}_t)$**-martingale** if it is both a super- and submartingale, i.e. $E[|X_t|] < \infty$ and $E[X_t|\mathcal{F}_s] = X_s$ a.s. for all $s \leq t$.

Definition. For a stochastic process $X = (X_t)_{t \in T}$, the **natural filtration** is

$$\mathcal{F}_t^X := \sigma(\{X_s : s \leq t\}).$$

We say X is a supermartingale (resp. submartingale / martingale) if it is an $(\mathcal{F}_t^X)$-supermartingale (resp. submartingale / martingale).

Exercise. If $X = (X_t)_{t \in T}$ is a $(\mathcal{F}_t)$-supermartingale, show that X is a supermartingale with respect to its natural filtration.

Example. (1) Let $(Y_n)_{n \in \mathbb{N}}$ be integrable, i.i.d. random variables and $S_n = \sum_{j=1}^{n} Y_j$. Then S_n is a supermartingale (resp. submartingale, martingale) if $E[Y_1] \leq 0$ (resp. ≥ 0, $= 0$).

(2) Let $(Y_n)_{n \in \mathbb{N}}$ be i.i.d. with $P(Y_n = 0) = P(Y_n = 2) = 1/2$. Let $X_0 = 1$ and $X_n = X_0 \prod_{j=1}^{n} Y_j$. Then

$$E[X_{n+1}|\mathcal{F}_n] = E[X_n Y_{n+1}|\mathcal{F}_n] = X_n E[Y_{n+1}|\mathcal{F}_n] = X_n E[Y_{n+1}] = X_n \cdot 1 = X_n$$

(using Proposition 1 and Example 1), so (X_n) is a martingale.

(3) Let $Z \in L^1(P)$ and $(\mathcal{F}_n)$ a filtration. Then $X_n := E[Z|\mathcal{F}_n]$ is a martingale. (Exercise: verify using the tower property.)

Definition (Predictable processes and martingale transform). A process $(H_n)_{n\geq 1}$ is $(\mathcal{F}_n)$**-predictable** if H_n is $\mathcal{F}_{n-1}$-measurable for all $n \in \mathbb{N}$.
If $(H_n)_{n\in\mathbb{N}}$ and $(X_n)_{n\in\mathbb{Z}_{\geq 0}}$ are stochastic processes, define the **martingale transform** (discrete stochastic integral) by

$$(H \cdot X)_n = \begin{cases} \sum_{k=1}^{n} H_k(X_k - X_{k-1}) & n \in \mathbb{N}, \\ 0 & n = 0. \end{cases}$$

The following result shows there is no gambling strategy that makes an unfavourable game favourable: if (H_n) is the bet placed at time n (decided using information up to time $n-1$) and (X_n) is the player's total winnings, then a non-negative bounded betting strategy (H_n) cannot improve a supermartingale (unfavourable game).

Theorem. Let $(H_n)_{n\geq 1}$ be a non-negative, bounded, $(\mathcal{F}_n)$-predictable process. If (X_n) is an $(\mathcal{F}_n)$-supermartingale (resp. martingale or submartingale), then so is $(H \cdot X)$.

Proof. For all $n \in \mathbb{Z}_{\geq 0}$, using Proposition 1 (since H_{n+1} is $\mathcal{F}_n$-measurable):

$$\begin{aligned} E[(H \cdot X)_{n+1}|\mathcal{F}_n] - (H \cdot X)_n &= E[H_{n+1}(X_{n+1} - X_n)|\mathcal{F}_n] \\ &= H_{n+1}\, E[X_{n+1} - X_n|\mathcal{F}_n] \\ &= H_{n+1}(E[X_{n+1}|\mathcal{F}_n] - X_n). \end{aligned}$$

Since (X_n) is a supermartingale, $E[X_{n+1}|\mathcal{F}_n] - X_n \leq 0$ a.s., and since $H_{n+1} \geq 0$, we get $E[(H \cdot X)_{n+1}|\mathcal{F}_n] \leq (H \cdot X)_n$ a.s. □

Definition (Stopping time). A map $T : \Omega \to \mathbb{Z}_{\geq 0} \cup \{\infty\}$ is an $(\mathcal{F}_n)$**-stopping time** if $\{T \leq n\} \in \mathcal{F}_n$ for all $n \in \mathbb{Z}_{\geq 0}$.

Exercise. Show that $\{T \leq n\} \in \mathcal{F}_n$ for all n if and only if $\{T = n\} \in \mathcal{F}_n$ for all n.

Corollary. Let T be an $(\mathcal{F}_n)$-stopping time. If (X_n) is an $(\mathcal{F}_n)$-supermartingale (resp. submartingale / martingale), then so is $X_n^T := X_{n\wedge T}$.

Proof. Set $H_n = \mathbf{1}_{\{T\geq n\}}$. Then $\{T \geq n\} = \{T \leq n-1\}^c \in \mathcal{F}_{n-1}$ for all $n \in \mathbb{N}$, so (H_n) is $(\mathcal{F}_n)$-predictable. It is also bounded and non-negative. Moreover,

$$(H \cdot X)_n = \sum_{k=1}^{n} \mathbf{1}_{\{T\geq k\}}(X_k - X_{k-1}) = \sum_{k=1}^{n\wedge T}(X_k - X_{k-1}) = X_{n\wedge T} - X_0.$$

By Theorem 2, $X_{n\wedge T} - X_0 = (H \cdot X)_n$ is a supermartingale, and hence so is $X_{n\wedge T}$ (adding the constant X_0). □

A stopping time T is **bounded** if there exists $N \in \mathbb{N}$ with $P(T \leq N) = 1$.

Corollary. Let T_1, T_2 be bounded $(\mathcal{F}_n)$-stopping times with $T_1 \leq T_2$ a.s. If (X_n) is an $(\mathcal{F}_n)$-submartingale, then $E[X_{T_1}] \leq E[X_{T_2}]$.

Proof. Let N be such that $P(T_1 \leq T_2 \leq N) = 1$. Define $H_n = \mathbf{1}_{\{T_2 \geq n\}} - \mathbf{1}_{\{T_1 \geq n\}} = \mathbf{1}_{\{T_1 < n \leq T_2\}}$. This is non-negative, bounded, and $(\mathcal{F}_n)$-predictable. Moreover,

$$(H \cdot X)_N = \sum_{k=1}^{N} H_k(X_k - X_{k-1}) = X_{T_2} - X_{T_1}.$$

By Theorem 2, $E[X_{T_2} - X_{T_1}] = E[(H \cdot X)_N] \geq E[(H \cdot X)_0] = 0$. □

Definition (Upcrossings). For a sequence $(X_n)_{n \in \mathbb{Z}_{\geq 0}}$ and $a < b$, the number of **upcrossings of** $[a, b]$ by $(X_n)_{n \in \mathbb{Z}_{\geq 0}}$ is the supremum over all k such that there exist $0 \leq m_1 < n_1 < \cdots < m_k < n_k$ with $X_{m_i} \leq a$ and $X_{n_i} \geq b$ for all $1 \leq i \leq k$.
A sequence (X_n) has a limit in $[-\infty, \infty]$ if and only if the number of upcrossings of $[a, b]$ is finite for every pair of rationals $a < b$.

Lemma (Doob's upcrossing lemma). Let $(Y_n)_{n \geq 0}$ be a submartingale and $U_N^{a,b}$ denote the number of upcrossings of $[a, b]$ by $(Y_{n \wedge N})$. Then

$$E[U_N^{a,b}] \leq \frac{1}{b-a}\Big(E[(Y_N - a)^+] - E[(Y_0 - a)^+]\Big) \leq \frac{1}{b-a} E[(Y_N - a)^+].$$

Proof. Since $\varphi(t) = \dfrac{(t-a)^+}{b-a}$ is convex and non-decreasing, by conditional Jensen (Corollary 1), $Z_n := \varphi(Y_n) = \dfrac{(Y_n - a)^+}{b-a}$ is a $(\mathcal{F}_n)$-submartingale.
Define $T_0 = 0$ and inductively for $k \in \mathbb{N}$:

$$S_k = \inf\{n \geq T_{k-1} : Y_n \leq a\} \wedge N, \qquad T_k = \inf\{n \geq S_k : Y_n \geq b\} \wedge N.$$

By induction, each S_k and T_k is an $(\mathcal{F}_n)$-stopping time. Note that $U_N^{a,b} \leq$ the number of complete upcrossings encoded by the T_k, and one can verify:

$$U_N^{a,b} \leq \sum_{k=1}^{N} (Z_{T_k} - Z_{S_k}).$$

Rearranging and using that $Z_N - Z_0 \geq \sum_{k=1}^{N} (Z_{T_k} - Z_{S_k}) - \sum_{k=1}^{N} (Z_{S_k} - Z_{T_{k-1}})$:

$$E[U_N^{a,b}] \leq E[Z_N - Z_0] - \sum_{k=1}^{N} E[Z_{S_k} - Z_{T_{k-1}}].$$

By Corollary 2 applied to the submartingale (Z_n) and the bounded stopping times $T_{k-1} \leq S_k$, we have $E[Z_{S_k} - Z_{T_{k-1}}] \geq 0$. Hence

$$E[U_N^{a,b}] \leq E[Z_N - Z_0] = \frac{1}{b-a}\Big(E[(Y_N - a)^+] - E[(Y_0 - a)^+]\Big).$$ □

If (X_n) is a submartingale, the expected number of upcrossings $U_\infty^{a,b}$ satisfies, by the monotone convergence theorem:

$$E[U_\infty^{a,b}] = \lim_{N\to\infty} E[U_N^{a,b}] \le \sup_{N\in\mathbb{N}} \frac{1}{b-a} E[(X_N - a)^+]. \tag{2.1}$$

Theorem (Martingale convergence theorem). Let $(X_n)_{n\in\mathbb{Z}_{\ge 0}}$ be a submartingale with $\sup_{n\in\mathbb{N}} E[X_n^+] < \infty$. Then $X_n \xrightarrow{\text{a.s.}} X$ as $n \to \infty$ for some $X \in L^1(P)$. In particular, every non-negative supermartingale converges a.s.

Proof. Since (X_n) is a submartingale, $E[X_n] \ge E[X_0]$, so $E[X_n^+] = E[X_n] + E[X_n^-] \le E[X_n] + E[X_0^-]$ (wait, let us be more careful). We have $E[X_n^-] = E[X_n^+] - E[X_n] \le E[X_n^+] - E[X_0]$, so

$$\sup_n E[X_n^-] \le \sup_n E[X_n^+] - E[X_0] < \infty.$$

For all $a < b \in \mathbb{Q}$, by (2.1): $E[U_\infty^{a,b}] \le \dfrac{1}{b-a} \sup_n E[(X_n - a)^+] \le \dfrac{1}{b-a}(\sup_n E[X_n^+] + |a|) < \infty$, hence $U_\infty^{a,b} < \infty$ a.s. Therefore

$$P\left(\limsup_{n\to\infty} X_n \ne \liminf_{n\to\infty} X_n\right) = P\left(\bigcup_{\substack{a,b\in\mathbb{Q}\\ a<b}} \{U_\infty^{a,b} = \infty\}\right) = 0.$$

Thus $X := \lim_{n\to\infty} X_n$ exists a.s. in $[-\infty, \infty]$. By Fatou's lemma:

$$E[X^+] \le \liminf_{n\to\infty} E[X_n^+] \le \sup_n E[X_n^+] < \infty, \quad E[X^-] \le \liminf_{n\to\infty} E[X_n^-] \le \sup_n E[X_n^-] < \infty.$$

Hence $X \in L^1(P)$, in particular X is finite a.s. □

The assumption $\sup_n E[X_n^+] < \infty$ is necessary: the symmetric simple random walk on $\mathbb{Z}$ satisfies $\liminf X_n = -\infty$ and $\limsup X_n = +\infty$ a.s.

Corollary. Let (X_n) be an $(\mathcal{F}_n)$-martingale with bounded increments: $\exists\, M > 0$ such that $\sup_{n\ge 0} |X_{n+1} - X_n| \le M$ a.s. Then with probability one, either:

(a) $\liminf_{n\to\infty} X_n = -\infty$ and $\limsup_{n\to\infty} X_n = +\infty$, or

(b) $\lim_{n\to\infty} X_n$ exists and is finite.

Proof. By replacing X_n with $X_n - X_0$, assume $X_0 = 0$. For $K > 0$, let $N_K = \inf\{n : X_n \le -K\}$, a stopping time. Then $X_{n\wedge N_K} \le K + M$ for all n, so $\sup_n E[X_{n\wedge N_K}^+] \le K + M < \infty$. By the MCT (Theorem 2), $\lim_{n\to\infty} X_{n\wedge N_K}$ exists and is finite a.s.

On the event $\{N_K = \infty\}$ (i.e. $X_n > -K$ for all n), $X_n = X_{n\wedge N_K} \to$ a finite limit. Hence on $\bigcap_{K\in\mathbb{N}} \{N_K = \infty\} = \{\liminf_n X_n > -\infty\}$, the limit exists and is finite. By replacing X_n with $-X_n$ and repeating, on $\{\limsup_n X_n < \infty\}$ the limit exists and is finite. Denoting $C = \{\lim_n X_n \text{ exists and is finite}\}$:

$$\{\limsup X_n < \infty\} \cup \{\liminf X_n > -\infty\} \subseteq C.$$

Hence $C^c \subseteq \{\limsup X_n = \infty\} \cap \{\liminf X_n = -\infty\}$. Combined with $C \cup C^c = \Omega$, this gives the dichotomy. □

Corollary (Second Borel–Cantelli lemma). Let $(\mathcal{F}_n)_{n\geq 0}$ be a filtration and $A_n \in \mathcal{F}_n$ for $n \in \mathbb{N}$. Then, up to a P-null set:

$$\{A_n \text{ i.o.}\} = \left\{\omega : \sum_{n=1}^{\infty} P(A_n|\mathcal{F}_{n-1}) = \infty\right\}.$$

Proof. Note that $M_n = \sum_{k=1}^{n}(\mathbf{1}_{A_k} - P(A_k|\mathcal{F}_{k-1}))$ is a martingale with $|M_n - M_{n-1}| \leq 1$. By Corollary 2, $P(C) = 1$ where $C \cup D = \Omega$ a.s., $C = \{\lim M_n \text{ exists and is finite}\}$, and $D = \{\limsup M_n = +\infty, \liminf M_n = -\infty\}$.
On C: $\sum \mathbf{1}_{A_k}$ converges iff $\sum P(A_k|\mathcal{F}_{k-1})$ converges (their difference M_n converges), i.e. $\{A_n \text{ i.o.}\} \cap C = \{\sum P(A_k|\mathcal{F}_{k-1}) = \infty\} \cap C$.
On D: both $\sum \mathbf{1}_{A_k} = \infty$ and $\sum P(A_k|\mathcal{F}_{k-1}) = \infty$. So the equality holds on $C \cup D = \Omega$ a.s. □

2.1 Uniform Integrability and Martingales

Let (X_n) be an L^1-bounded $(\mathcal{F}_n)$-martingale ($\sup_n E[|X_n|] < \infty$). By Theorem 2, $X_n \to X$ a.s. with $X \in L^1$. For $N > n$, $E[X_N|\mathcal{F}_n] = X_n$ a.s. We would like to let $N \to \infty$ and conclude $E[X|\mathcal{F}_n] = X_n$, but this requires interchanging a limit with conditional expectation, which is not always valid (see Example 2(2)). The correct condition is uniform integrability.

Definition (Uniform integrability). A collection of random variables $\{X_i : i \in I\}$ is **uniformly integrable (u.i.)** if

$$\lim_{M\to\infty} \sup_{i\in I} E\big[|X_i|\, \mathbf{1}_{\{|X_i|>M\}}\big] = 0.$$

Remark. (a) If $\{X_i\}$ is u.i., then $\{X_i\}$ is L^1-bounded. (Pick M large enough that $\sup_i E[|X_i|\mathbf{1}_{\{|X_i|>M\}}] \leq 1$; then $E[|X_i|] \leq M + 1$.)

(b) If $\{X_i\}$ is L^p-bounded for some $p > 1$ (i.e. $\sup_i E[|X_i|^p] < \infty$), then $\{X_i\}$ is u.i. (Use $|X|\mathbf{1}_{\{|X|>M\}} \leq M^{1-p}|X|^p$.)

(c) (Exercise) If $|X_i| \leq Z$ a.s. for all i for some $Z \in L^1(P)$, then $\{X_i\}$ is u.i. (Hint: $|X|\mathbf{1}_{\{|X|>M\}} \leq Z\mathbf{1}_{\{Z>M\}} \to 0$ in L^1 by DCT.)

(d) (Exercise) The collection (X_n) from Example 2(2) is not u.i.

(e) (Exercise) If $\{X_i\}$ and $\{Y_i\}$ are u.i., so is $\{|X_i| + |Y_i|\}$.

Theorem. Assume $X_n \to X$ in probability. Then the following are equivalent:

(a) $\{X_n : n \in \mathbb{N}\}$ is u.i.

(b) $X_n \to X$ in L^1, i.e. $\lim_{n\to\infty} E[|X_n - X|] = 0$, with $X_n, X \in L^1$.

(c) $\lim_{n\to\infty} E[|X_n|] = E[|X|]$.

Proof. See Durrett, Theorem 4.6.3. Note that by Remark 2.1(c), this generalises the dominated convergence theorem. □

Lemma. Let $Z \in L^1(P)$ and $\mathcal{F}_i \overset{\sigma}{\subset} \mathcal{F}$ for each $i \in I$. Then $\{E[Z|\mathcal{F}_i] : i \in I\}$ is u.i.

Proof. We use the following fact: for any $Z \in L^1(P)$ and $\varepsilon > 0$, there exists $\delta > 0$ such that

$$P(A) < \delta \Rightarrow \int_A |Z| \,\mathrm{d}P < \varepsilon. \tag{$*$}$$

(Proof of $(*)$: by contradiction, suppose there exist $\varepsilon > 0$ and events A_n with $P(A_n) \leq n^{-1}$ and $\int_{A_n} |Z| \,\mathrm{d}P \geq \varepsilon$. Then $|Z|\mathbf{1}_{A_n} \to 0$ a.s. (along a subsequence), and $|Z|\mathbf{1}_{A_n} \leq |Z| \in L^1$, so by DCT $\int |Z|\mathbf{1}_{A_n} \,\mathrm{d}P \to 0$, a contradiction.)

Fix $\varepsilon > 0$ and choose $\delta > 0$ by $(*)$ for Z. Choose M large enough that $E[|Z|]/M < \delta$. By Markov's inequality:

$$P(|E[Z|\mathcal{F}_i]| > M) \leq \frac{E[|E[Z|\mathcal{F}_i]|]}{M} \leq \frac{E[E[|Z||\mathcal{F}_i]]}{M} = \frac{E[|Z|]}{M} < \delta.$$

By conditional Jensen and the integral characterisation of conditional expectation:

$$\int_{\{|E[Z|\mathcal{F}_i]|>M\}} |E[Z|\mathcal{F}_i]| \,\mathrm{d}P \leq \int_{\{|E[Z|\mathcal{F}_i]|>M\}} E[|Z|\big|\mathcal{F}_i] \,\mathrm{d}P = \int_{\{|E[Z|\mathcal{F}_i]|>M\}} |Z| \,\mathrm{d}P \overset{(*)}{<} \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, $\{E[Z|\mathcal{F}_i]\}$ is u.i. □

We now recall two powerful tools.

Definition. Let Ω be a set.

(1) A collection $\mathcal{P}$ of subsets of Ω is a **π-system** if it is closed under finite intersections: $A, B \in \mathcal{P} \Rightarrow A \cap B \in \mathcal{P}$.

(2) A collection $\mathcal{L}$ of subsets of Ω is a **λ-system** if:

   (a) $\Omega \in \mathcal{L}$,

   (b) $A, B \in \mathcal{L}$, $A \subset B$ implies $B \setminus A \in \mathcal{L}$,

   (c) $A_n \in \mathcal{L}$, $A_n \uparrow A$ implies $A \in \mathcal{L}$.

Theorem (π-λ theorem (Dynkin)). If $\mathcal{P}$ is a π-system and $\mathcal{L}$ is a λ-system with $\mathcal{P} \subset \mathcal{L}$, then $\sigma(\mathcal{P}) \subset \mathcal{L}$.

Theorem (Functional monotone class theorem (FMCT)). Let $\mathcal{H}$ be a collection of real-valued functions and $\mathcal{A}$ a π-system with $\Omega \in \mathcal{A}$ such that:

(i) $A \in \mathcal{A} \Rightarrow \mathbf{1}_A \in \mathcal{H}$,

(ii) $\mathcal{H}$ is a vector space (closed under addition and scalar multiplication),

(iii) If $f_n \in \mathcal{H}$, $f_n \geq 0$, $f_n \uparrow f$ with f bounded, then $f \in \mathcal{H}$.

Then $\mathcal{H} \supset b\sigma(\mathcal{A})$ (the bounded $\sigma(\mathcal{A})$-measurable functions).

Exercise. (i) Let P, Q be probability measures on $(\Omega, \sigma(\mathcal{P}))$ for a π-system $\mathcal{P}$. If $P(A) = Q(A)$ for all $A \in \mathcal{P}$, show $P = Q$ on $\sigma(\mathcal{P})$. (Hint: take $\mathcal{L} = \{A \in \mathcal{F} : P(A) = Q(A)\}$.)

(ii) Let X be independent of $\mathcal{E} \overset{\sigma}{\subset} \mathcal{F}$, and $\mathcal{D} \overset{\sigma}{\subset} \mathcal{F}$. Show $E[X|\sigma(\mathcal{D} \cup \mathcal{E})] = E[X|\mathcal{D}]$.

Notation. We write $\mathcal{F}_n \uparrow \mathcal{F}_\infty$ to mean $\mathcal{F}_\infty = \sigma\left(\bigcup_n \mathcal{F}_n\right)$.

Theorem (L^1 characterisation of u.i. martingales). Let $(\mathcal{F}_n)$ be a filtration with $\mathcal{F}_n \uparrow \mathcal{F}_\infty$. Then (X_n) is a u.i. $(\mathcal{F}_n)$-martingale if and only if there exists $X \in L^1(P)$ such that $X_n = E[X|\mathcal{F}_n]$ a.s. for all n.
In this case:

(i) $X_n \to X_\infty := E[X|\mathcal{F}_\infty]$ a.s. and in L^1.

(ii) $X_n = E[X_\infty|\mathcal{F}_n]$ a.s. for all n.

Proof. ($\Rightarrow$) Suppose (X_n) is u.i. By Remark 2.1(a), (X_n) is L^1-bounded. By Theorem 2, $X_n \to X_\infty$ a.s., and $X_\infty \in L^1$. Since (X_n) is u.i. and converges a.s., by Theorem 2.1, $X_n \to X_\infty$ in L^1.
For any $A \in \mathcal{F}_n$ and $N > n$: $\displaystyle\int_A X_N \,\mathrm{d}P = \int_A X_n \,\mathrm{d}P$ (martingale property). Let $N \to \infty$; since $X_N \to X_\infty$ in L^1: $\displaystyle\int_A X_\infty \,\mathrm{d}P = \int_A X_n \,\mathrm{d}P$ for all $A \in \mathcal{F}_n$.

Therefore $X_n = E[X_\infty|\mathcal{F}_n]$ a.s. This proves (ii) with $X = X_\infty$, and it remains to show $X_\infty = E[X|\mathcal{F}_\infty]$ for any $X \in L^1$ with $X_n = E[X|\mathcal{F}_n]$: define

$$\mathcal{L} = \left\{ A \in \mathcal{F} : \int_A X_\infty \, \mathrm{d}P = \int_A X \, \mathrm{d}P \right\}.$$

We verify $\mathcal{L}$ is a λ-system (using DCT for condition (c)) containing the π-system $\bigcup_n \mathcal{F}_n$. Since $\int_A X_\infty \, \mathrm{d}P = \int_A X_n \, \mathrm{d}P = \int_A X \, \mathrm{d}P$ for all $A \in \mathcal{F}_n$. By the π-λ theorem, $\mathcal{F}_\infty \subset \mathcal{L}$, so $X_\infty = E[X|\mathcal{F}_\infty]$ a.s.
($\Leftarrow$) If $X_n = E[X|\mathcal{F}_n]$, then by Lemma 2.1, (X_n) is u.i. By the tower property, (X_n) is a martingale. $\square$

Remark (Uniformly integrable submartingales). Let (X_n) be a u.i. $(\mathcal{F}_n)$-submartingale. Then $X_n \to X_\infty$ a.s. and in L^1, and $X_n \leq E[X_\infty|\mathcal{F}_n]$ a.s. for all n.

Proof. L^1-boundedness and a.s. convergence follow as before. For the inequality: for $N > n$, $X_n \leq E[X_N|\mathcal{F}_n]$ a.s. (submartingale). Since $X_N \to X_\infty$ in L^1, $E[X_N|\mathcal{F}_n] \xrightarrow{L^1} E[X_\infty|\mathcal{F}_n]$, and hence $X_n \leq E[X_\infty|\mathcal{F}_n]$ a.s. $\square$

2.2 Backwards Martingales

Let $\mathbb{Z}_{<0} = \{-1, -2, \ldots\}$ denote the negative integers.

Definition (Backwards martingale). A **backwards martingale** is a martingale indexed by $\mathbb{Z}_{<0}$. That is, $(X_n)_{n \leq -1}$ is adapted to a filtration $(\mathcal{F}_n)_{n \leq -1}$ (with $\mathcal{F}_m \overset{\sigma}{\subset} \mathcal{F}_n$ for $m \leq n$) and

$$E[X_{n+1}|\mathcal{F}_n] = X_n \text{ a.s.} \quad \text{for all } n \leq -2.$$

Example. Let $(\xi_i)_{i \geq 1}$ be i.i.d. with $E[|\xi_1|] < \infty$. For $n \in \mathbb{N}$, let $S_n = \sum_{i=1}^n \xi_i$, $X_{-n} = S_n/n$, and $\mathcal{F}_{-n} = \sigma(S_n, S_{n+1}, \ldots) = \sigma(S_n, \xi_{n+1}, \xi_{n+2}, \ldots)$. We claim $(X_{-n})_{n \geq 1}$ is an $(\mathcal{F}_{-n})$-backwards martingale.
For $j, k \leq n-1$, since ξ_j and ξ_k are exchangeable given $\mathcal{F}_{-(n-1)} = \sigma(S_{n-1}, \xi_n, \xi_{n+1}, \ldots)$, we have $E[\xi_j|\mathcal{F}_{-(n-1)}] = E[\xi_k|\mathcal{F}_{-(n-1)}]$. Therefore:

$$E[\xi_n|\mathcal{F}_{-(n-1)}] = \frac{1}{n-1} \sum_{k=1}^{n-1} E[\xi_k|\mathcal{F}_{-(n-1)}] = E\left[\frac{S_{n-1}}{n-1} \,\middle|\, \mathcal{F}_{-(n-1)} \right] = \frac{S_{n-1}}{n-1} = X_{-(n-1)}.$$

Since $X_{-n} = \dfrac{S_{n-1} + \xi_n}{n}$:

$$E[X_{-n}|\mathcal{F}_{-(n-1)}] = \frac{1}{n}\big(E[S_{n-1}|\mathcal{F}_{-(n-1)}] + E[\xi_n|\mathcal{F}_{-(n-1)}]\big) = \frac{1}{n}\left(S_{n-1} + \frac{S_{n-1}}{n-1} \right) = \frac{S_{n-1}}{n-1} = X_{-(n-1)}.$$

Theorem (Backwards martingale convergence theorem). Let $(X_n)_{n\le -1}$ be an $(\mathcal{F}_n)$-backwards martingale and let $\mathcal{F}_{-\infty} = \bigcap_{n\le -1} \mathcal{F}_n$. Then $X_n \to E[X_{-1}|\mathcal{F}_{-\infty}]$ a.s. and in L^1 as $n \to -\infty$.

Proof. **Step 1: Almost sure convergence.** Fix $N \in \mathbb{N}$ and define

$$Y_n = \begin{cases} X_{n-N+1} & 0 \le n \le N, \\ X_{-1} & n \ge N+1. \end{cases}$$

Then (Y_n) is a martingale. Let $U_N^{a,b}$ be the number of upcrossings of $[a,b]$ by $(X_{-N}, X_{-(N-1)}, \ldots, X_{-1})$. This equals the number of upcrossings of $[a,b]$ by $(Y_0, Y_1, \ldots, Y_N)$, hence by the upcrossing lemma:

$$E[U_N^{a,b}] \le \frac{1}{b-a} E[(X_{-1}-a)^+] < \infty.$$

Note $U_N^{a,b}$ counts upcrossings of $(X_{-N}, \ldots, X_{-1})$, which equals the number of *down*crossings of $(X_{-1}, X_{-2}, \ldots, X_{-N})$. Let $N \to \infty$; by MCT, the expected total number of downcrossings of $[a,b]$ by $(X_{-1}, X_{-2}, \ldots)$ is finite. Hence for all $a < b \in \mathbb{Q}$, the number of downcrossings is a.s. finite, so $\lim_{n\to-\infty} X_n$ exists a.s. and belongs to $[-\infty, \infty]$; call it X.

Step 2: L^1 convergence. By the martingale property, $X_n = E[X_{-1}|\mathcal{F}_n]$ for all $n \le -1$. Hence by Lemma 2.1, $(X_n)_{n\le -1}$ is u.i. Since $X_n \to X$ a.s., Theorem 2.1 gives $X_n \to X$ in L^1.

Step 3: Identification of limit. Since $X = \liminf_{n\to-\infty} X_n$ is $\mathcal{F}_k$-measurable for all $k \le -1$ (each X_n with $n \le k$ is $\mathcal{F}_k$-measurable), X is $\mathcal{F}_{-\infty}$-measurable. By conditional Jensen and L^1-convergence:

$$E[X_n|\mathcal{F}_{-\infty}] \xrightarrow{L^1} E[X|\mathcal{F}_{-\infty}] = X.$$

By the tower property, for all $n \le -1$: $E[X_n|\mathcal{F}_{-\infty}] = E[E[X_{-1}|\mathcal{F}_n]|\mathcal{F}_{-\infty}] = E[X_{-1}|\mathcal{F}_{-\infty}]$. Hence $X = E[X_{-1}|\mathcal{F}_{-\infty}]$. □

Corollary (Strong law of large numbers). Let (ξ_i) be i.i.d. with mean μ. Then $S_n/n \to \mu$ a.s. and in L^1 as $n \to \infty$.

Proof. By Example 2.2 and Theorem 2.2, $S_n/n \to E[\xi_1|\mathcal{F}_{-\infty}]$ a.s. and in L^1, where $\mathcal{F}_{-\infty} = \bigcap_n \sigma(S_n, S_{n+1}, \ldots)$ is the tail σ-field. Any event in $\mathcal{F}_{-\infty}$ is a symmetric event for the i.i.d. sequence (ξ_n); by the Hewitt–Savage 0-1 law (see Durrett, Theorem 2.5.4), every such event has probability 0 or 1. Hence $\mathcal{F}_{-\infty}$ is P-trivial, so $E[\xi_1|\mathcal{F}_{-\infty}] = E[\xi_1] = \mu$ a.s. □

2.3 Maximal Inequalities

Exercise. If S, T are $(\mathcal{F}_n)$-stopping times, show $S \wedge T$ and $S \vee T$ are also $(\mathcal{F}_n)$-stopping times.

Definition (Maximal function). For a stochastic process $X = (X_n)_{n\in\mathbb{Z}_{\geq 0}}$, define

$$X_n^* = \sup_{0\leq k\leq n} |X_k|, \quad X^* = \sup_{n\in\mathbb{Z}_{\geq 0}} |X_n|.$$

Also define $\|X\|_p := \sup_n (E[|X_n|^p])^{1/p}$ and $\|X^*\|_p := (E[|X^*|^p])^{1/p}$.

Theorem (Doob's weak L^1 inequality). Let $(X_n)_{n\in\mathbb{Z}_{\geq 0}}$ be a non-negative $(\mathcal{F}_n)$-submartingale. Then for all $\lambda > 0$:

(i) $P(X_n^* \geq \lambda) \leq \dfrac{1}{\lambda}\displaystyle\int_{\{X_n^*\geq\lambda\}} X_n \,\mathrm{d}P \leq \dfrac{E[X_n]}{\lambda}$.

(ii) $P(X^* \geq \lambda) \leq \dfrac{1}{\lambda}\sup_n E[X_n] = \dfrac{\|X\|_1}{\lambda}$.

We prepare with a lemma.

Lemma. Let (X_n) be an $(\mathcal{F}_n)$-submartingale and T an $(\mathcal{F}_n)$-stopping time with $T \leq n$ a.s. Then $E[X_T] \leq E[X_n]$.

Proof.

$$E[X_T] = \sum_{k=0}^{n}\int_{\{T=k\}} X_T \,\mathrm{d}P = \sum_{k=0}^{n}\int_{\{T=k\}} X_k \,\mathrm{d}P \leq \sum_{k=0}^{n}\int_{\{T=k\}} E[X_n|\mathcal{F}_k] \,\mathrm{d}P = \sum_{k=0}^{n}\int_{\{T=k\}} X_n \,\mathrm{d}P = E[X_n],$$

using the submartingale property and that $\{T = k\} \in \mathcal{F}_k$. □

Proof of Theorem 2.3. (i) Let $T = \inf\{n \geq 0 : X_n \geq \lambda\}$ (with $T = \infty$ if no such n exists). T is a stopping time. Then $\{T \leq n\} = \{X_n^* \geq \lambda\}$, and on $\{T \leq n\}$, $X_T \geq \lambda$. Thus:

$$\begin{aligned}\lambda P(X_n^* \geq \lambda) = \lambda P(T \leq n) &\leq \int_{\{T\leq n\}} X_T \,\mathrm{d}P = \int_{\{T\leq n\}} X_{T\wedge n} \,\mathrm{d}P \\ &= E[X_{T\wedge n}] - \int_{\{T>n\}} X_n \,\mathrm{d}P \leq E[X_n] - \int_{\{T>n\}} X_n \,\mathrm{d}P = \int_{\{T\leq n\}} X_n \,\mathrm{d}P,\end{aligned}$$

where we used Lemma 2.3 and the fact that $X_{T\wedge n}\mathbf{1}_{\{T\leq n\}} = X_T\mathbf{1}_{\{T\leq n\}}$. Since $\{T \leq n\} = \{X_n^* \geq \lambda\}$, we obtain $P(X_n^* \geq \lambda) \leq \dfrac{1}{\lambda}\displaystyle\int_{\{X_n^*\geq\lambda\}} X_n \,\mathrm{d}P \leq \dfrac{E[X_n]}{\lambda}$.

(ii) For $\varepsilon \in (0, \lambda)$: $\{X^* \geq \lambda\} \subset \bigcup_n \{X_n^* \geq \lambda - \varepsilon\}$. Letting $n \to \infty$ in (i):

$$P(X^* \geq \lambda) \leq \lim_{n\to\infty} P(X_n^* \geq \lambda - \varepsilon) \leq \frac{\sup_n E[X_n]}{\lambda - \varepsilon}.$$

Let $\varepsilon \downarrow 0$. □

Theorem (Doob's L^p maximal inequality). Let $(X_n)_{n\in\mathbb{Z}_{\geq 0}}$ be a non-negative $(\mathcal{F}_n)$-submartingale and $p \in (1, \infty)$, $q = p/(p-1)$. Then:

(a) $\|X_n^*\|_p \le q\|X_n\|_p$.

(b) $\|X^*\|_p \le q\|X\|_p$.

Proof. (b) follows from (a) by letting $n \to \infty$ via MCT. For (a), it suffices to prove the following general claim: if non-negative r.v.'s Y, Z satisfy

$$P(Z \ge \lambda) \le \frac{1}{\lambda}\int_{\{Z\ge\lambda\}} Y \,\mathrm{d}P \quad \text{for all } \lambda > 0, \tag{$**$}$$

then $\|Z\|_p \le q\|Y\|_p$. Apply this with $Z = X_n^*$, $Y = X_n$ (valid by Theorem 2.3(i)).
Case 1: $\|Z\|_p < \infty$**.** By Fubini and $(**)$:

$$\begin{aligned} E[Z^p] &= \int_0^\infty p\lambda^{p-1}P(Z \ge \lambda)\,\mathrm{d}\lambda \le \int_0^\infty p\lambda^{p-2}\int_{\{Z\ge\lambda\}} Y\,\mathrm{d}P\,\mathrm{d}\lambda \\ &= \int_\Omega Y\int_0^Z p\lambda^{p-2}\,\mathrm{d}\lambda\,\mathrm{d}P = \frac{p}{p-1}\int Y Z^{p-1}\,\mathrm{d}P \le \frac{p}{p-1}\|Y\|_p\|Z^{p-1}\|_q = q\|Y\|_p\,\|Z\|_p^{p/q}, \end{aligned}$$

using Hölder's inequality and $p/q = p-1$. Dividing by $\|Z\|_p^{p/q} = \|Z\|_p^{p-1}$ gives $\|Z\|_p \le q\|Y\|_p$.
Case 2: $\|Z\|_p = \infty$**.** Replace Z by $Z \wedge M$ for $M \in \mathbb{N}$. Since $(**)$ still holds for $Z \wedge M$, Case 1 gives $\|Z \wedge M\|_p \le q\|Y\|_p$. Let $M \to \infty$ by MCT. □

Example. There is no L^1 maximal inequality: for the martingale (X_n) of Example 2(2), one can show $P(X^* = 2^k) = 2^{-(k+1)}$ for $k = 0, 1, 2, \ldots$, so $E[X^*] = \sum_{k=0}^{\infty} 2^k \cdot 2^{-(k+1)} = \infty$, while $\|X\|_1 = \sup_n E[X_n] = 1 < \infty$.

Theorem (Doob decomposition). Every $(\mathcal{F}_n)$-submartingale $(X_n)_{n\ge0}$ admits a unique decomposition $X_n = M_n + A_n$ where:

- (M_n) is an $(\mathcal{F}_n)$-martingale,
- (A_n) is non-decreasing ($A_{n+1} \ge A_n$ for all $n \ge 0$), $(\mathcal{F}_n)$-predictable, and $A_0 = 0$.

Proof. Let $\delta_n = X_n - X_{n-1}$ for $n \ge 1$. If $X_n = M_n + A_n$ as above, then for $n \ge 1$:

$$E[\delta_n|\mathcal{F}_{n-1}] = E[M_n - M_{n-1}|\mathcal{F}_{n-1}] + E[A_n - A_{n-1}|\mathcal{F}_{n-1}] = 0 + (A_n - A_{n-1}),$$

since (M_n) is a martingale and $A_n - A_{n-1}$ is $\mathcal{F}_{n-1}$-measurable. With $A_0 = 0$:

$$A_n = \sum_{k=1}^n (A_k - A_{k-1}) = \sum_{k=1}^n E[\delta_k|\mathcal{F}_{k-1}].$$

This proves uniqueness (since $M_n = X_n - A_n$ is determined), and also defines the explicit decomposition. To verify existence: since (X_n) is a submartingale, $E[\delta_k|\mathcal{F}_{k-1}] \ge 0$, so (A_n) is non-decreasing and predictable. It remains to check (M_n) is a martingale:

$$E[M_n|\mathcal{F}_{n-1}] = E[X_n|\mathcal{F}_{n-1}] - E[A_n|\mathcal{F}_{n-1}] = E[X_n|\mathcal{F}_{n-1}] - A_{n-1} - E[\delta_n|\mathcal{F}_{n-1}] = E[X_n|\mathcal{F}_{n-1}] - A_{n-1} - (E[X_n|\mathcal{F}$$

□

2.4 Optional Stopping Theorem

Definition. Let T be an $(\mathcal{F}_n)$-stopping time with values in $\mathbb{Z}_{\geq 0} \cup \{\infty\}$. The **stopped σ-field** is

$$\mathcal{F}_T := \{A \in \mathcal{F} : A \cap \{T \leq n\} \in \mathcal{F}_n \text{ for all } n \in \mathbb{Z}_{\geq 0}\}.$$

Exercise. Let T be a stopping time.

(i) $\mathcal{F}_T$ is a σ-field.

(ii) $\mathcal{F}_T = \{A \in \mathcal{F} : A \cap \{T = n\} \in \mathcal{F}_n \text{ for all } n\}$.

(iii) If T_1, T_2 are stopping times with $T_1 \leq T_2$ a.s., then $\mathcal{F}_{T_1} \subset \mathcal{F}_{T_2}$.

(iv) If $T = n$ a.s. for some n, then $\mathcal{F}_T = \mathcal{F}_n$.

(v) If (X_n) is $(\mathcal{F}_n)$-adapted, then $X_T \mathbf{1}_{\{T<\infty\}}$ is $\mathcal{F}_T$-measurable.

For the optional stopping theorem, we extend the u.i. submartingale to time ∞ by $X_\infty := \lim_{n\to\infty} X_n$ (which exists in L^1 by Remark 2.1).

Proposition. If (X_n) is a u.i. $(\mathcal{F}_n)$-submartingale and T is a stopping time, then $(X_{n\wedge T})$ is a u.i. $(\mathcal{F}_n)$-submartingale and $E[|X_T|] < \infty$.

Proof. By Remark 2.1, $X_n \xrightarrow{L^1} X_\infty$. By Corollary 2, $(X_{n\wedge T})$ is a submartingale.
To show $E[|X_T|] < \infty$: since $(|X_n|)$ is a submartingale (by conditional Jensen) and $X_{n\wedge T}$ is also a submartingale:

$$\begin{aligned} E[|X_{n\wedge T}|] &= E[X^+_{n\wedge T}] + E[X^-_{n\wedge T}] \leq E[X^+_{n\wedge T}] + E[X^-_{n\wedge T}] + E[X^-_0] \\ &\leq 2\sum_{k=0}^{n-1} \int_{\{T=k\}} X^+_k \,\mathrm{d}P + 2\int_{\{T\geq n\}} X^+_n \,\mathrm{d}P + E[X^-_0] \\ &\leq 2E[X^+_\infty] + E[X^-_0] < \infty, \end{aligned}$$

using that (X^+_n) is a submartingale and Remark 2.1. By Fatou's lemma, $E[|X_T|] \leq 2E[X^+_\infty] + E[X^-_0] < \infty$.
For uniform integrability of $(X_{n\wedge T})$: $|X_{n\wedge T}| \leq |X_n| + |X_T|$, so

$$|X_{n\wedge T}|\mathbf{1}_{\{|X_{n\wedge T}|>M\}} \leq 2|X_n|\mathbf{1}_{\{|X_n|>M/2\}} + 2|X_T|\mathbf{1}_{\{|X_T|>M/2\}}.$$

Taking expectation and supremum over n: both terms $\to 0$ as $M \to \infty$ since (X_n) is u.i. and $X_T \in L^1$. $\square$

Proposition. Let (X_n) be a u.i. $(\mathcal{F}_n)$-submartingale and $T_1 \leq T_2$ stopping times. Then $E[X_{T_1}] \leq E[X_{T_2}]$.

Proof. First handle the case $T_2 = \infty$. By Lemma 2.3, $E[X_{n\wedge T_1}] \leq E[X_n]$ for all n. As $n \to \infty$: $E[X_{n\wedge T_1}] \to E[X_{T_1}]$ (since $(X_{n\wedge T_1})$ is u.i. and $X_{n\wedge T_1} \to X_{T_1}$ a.s.) and $E[X_n] \to E[X_\infty]$. Hence $E[X_{T_1}] \leq E[X_\infty] = E[X_{T_2}]$.
For general T_2: apply the above to the u.i. submartingale $(Y_n) := (X_{n\wedge T_2})$, noting $X_{T_1} = Y_{T_1}$ (since $T_1 \leq T_2$) and $Y_\infty = X_{T_2}$. □

Theorem (Optional stopping theorem). Let (X_n) be a u.i. $(\mathcal{F}_n)$-submartingale (resp. martingale) and $T_1 \leq T_2 \leq \cdots$ a non-decreasing sequence of $(\mathcal{F}_n)$-stopping times. Then $(X_{T_k})_{k\in\mathbb{N}}$ is an $(\mathcal{F}_{T_k})$-submartingale (resp. martingale). In particular, $k \mapsto E[X_{T_k}]$ is non-decreasing (resp. constant).

Proof. It suffices to show $E[X_{T_2}|\mathcal{F}_{T_1}] \geq X_{T_1}$ a.s. Let $A \in \mathcal{F}_{T_1}$. Define

$$T_1'(\omega) = \begin{cases} T_1(\omega) & \omega \in A, \\ T_2(\omega) & \omega \notin A. \end{cases}$$

We verify T_1' is a stopping time: $\{T_1' \leq n\} = (\{T_1 \leq n\} \cap A) \cup (\{T_2 \leq n\} \cap A^c)$. Both $A \cap \{T_1 \leq n\} \in \mathcal{F}_n$ (since $A \in \mathcal{F}_{T_1}$) and $A^c \cap \{T_2 \leq n\} \in \mathcal{F}_n$ (since $A^c \in \mathcal{F}_{T_1} \subset \mathcal{F}_{T_2}$). So T_1' is a stopping time with $T_1' \leq T_2$. By Proposition 2.4:

$$E[X_{T_1'}] \leq E[X_{T_2}],$$

i.e. $\int_A X_{T_1}\,\mathrm{d}P + \int_{A^c} X_{T_2}\,\mathrm{d}P \leq \int_A X_{T_2}\,\mathrm{d}P + \int_{A^c} X_{T_2}\,\mathrm{d}P$, giving $\int_A X_{T_1}\,\mathrm{d}P \leq \int_A X_{T_2}\,\mathrm{d}P$ for all $A \in \mathcal{F}_{T_1}$.
Since X_{T_1} is $\mathcal{F}_{T_1}$-measurable (Exercise 2.4(v)), we conclude $E[X_{T_2}|\mathcal{F}_{T_1}] \geq X_{T_1}$ a.s. □

Remark. (X_{T_k}) is also a u.i. submartingale (see homework).

Corollary. Let (M_n) be a u.i. $(\mathcal{F}_n)$-martingale. Then:

(a) $M_T = E[M_\infty|\mathcal{F}_T]$ for any $(\mathcal{F}_n)$-stopping time T.

(b) $\{M_T : T \text{ an } (\mathcal{F}_n)\text{-stopping time}\}$ is uniformly integrable.

Proof. For (a): by Lemma 2.1, it suffices to show (a). Apply Theorem 2.4 with $T_1 = T$ and $T_k = \infty$ for $k \geq 2$, noting that $M_\infty = E[M|\mathcal{F}_\infty]$ for some $M \in L^1$. □

Theorem. Let (X_n) be an $(\mathcal{F}_n)$-martingale and T an $(\mathcal{F}_n)$-stopping time satisfying:

(i) There exists $B \in (0, \infty)$ such that $E[|X_{n+1} - X_n| \,|\, \mathcal{F}_n] \leq B$ a.s. for all n.

(ii) $E[T] < \infty$.

Then (X_n) is a u.i. martingale, $X_T \in L^1$, and $E[X_T] = E[X_0]$.

Proof. Let $d_k = X_k - X_{k-1}$ for $k \geq 1$. Then for all n:

$$|X_{n\wedge T}| \leq |X_0| + \sum_{k=1}^{T\wedge n} |d_k| \leq |X_0| + \sum_{k=1}^{\infty} |d_k| \mathbf{1}_{\{k\leq T\}} =: Z.$$

We bound $E[Z]$:

$$\begin{aligned} E[Z] &= E[|X_0|] + \sum_{k=1}^{\infty} E[|d_k|\mathbf{1}_{\{k\leq T\}}] = E[|X_0|] + \sum_{k=1}^{\infty} E[\mathbf{1}_{\{k\leq T\}} E[|d_k| \,|\, \mathcal{F}_{k-1}]] \\ &\leq E[|X_0|] + B\sum_{k=1}^{\infty} P(k \leq T) = E[|X_0|] + B\, E[T] < \infty, \end{aligned}$$

where the interchange of sum and expectation is justified by Fubini/MCT, and we used $\sum_{k=1}^{\infty} P(k \leq T) = E[T]$.
Since $|X_{n\wedge T}| \leq Z$ for all n and $E[Z] < \infty$, $(X_{n\wedge T})$ is u.i. by Remark 2.1(c). By Theorem 2.1, $X_T \in L^1$. By Corollary 2.4: $X_{n\wedge T} = E[X_T|\mathcal{F}_n]$ for all n, so setting $n = 0$: $E[X_0] = E[X_T]$. □

Corollary (Wald's identity). Let $(Y_j)_{j\geq 1}$ be i.i.d. with mean μ, $S_n = \sum_{j=1}^{n} Y_j$, and T an $(\mathcal{F}_n^Y)$-stopping time with $E[T] < \infty$. Then $S_T \in L^1$ and $E[S_T] = \mu E[T]$.

Proof. $M_n = S_n - n\mu$ is an $(\mathcal{F}_n^Y)$-martingale. Condition (i) of Theorem 2.4: $E[|M_{n+1} - M_n| \,|\, \mathcal{F}_n^Y] = E[|Y_{n+1} - \mu| \,|\, \mathcal{F}_n^Y] = E[|Y_1 - \mu|] =: B < \infty$ (by independence). Condition (ii): $E[T] < \infty$. So $E[M_T] = E[M_0] = 0$, i.e. $E[S_T] = \mu E[T]$. □

Exercise (Simple random walk). Let $S_n = \sum_{i=1}^{n} X_i$, $S_0 = 0$, where X_i are i.i.d. with $P(X_1 = 1) = P(X_1 = -1) = \frac{1}{2}$. Let $\mathcal{F}_n = \mathcal{F}_n^S$.

(a) Show (S_n) is a martingale.

(b) For $a \in \mathbb{Z}$, show $T_a = \inf\{k \geq 0 : S_k = a\}$ is a stopping time.

(c) For $a, b \in \mathbb{N}$, let $T = T_{-a} \wedge T_b$. Using the stopped process $(S_{n\wedge T})$, show $P(T < \infty) = 1$ and $P(T_{-a} < T_b) = \dfrac{b}{a+b}$.

(d) Using (c), show $P(T_c < \infty) = 1$ for any $c \in \mathbb{Z}$.

(e) Show $E[T] = ab$ where $T = T_{-a} \wedge T_b$. (Hint: show $S_n^2 - n$ is a martingale and apply OST.)

(f) For $\sigma \in \mathbb{R}$, show $M_n := e^{\sigma S_n}\left(\frac{2}{e^{\sigma}+e^{-\sigma}}\right)^n$ is a martingale.

(g) Using (f) and the OST, show $E[s^{T_1}] = \dfrac{1-\sqrt{1-s^2}}{s}$ for $s \in (0,1)$.

(h) Expand $\sqrt{1-s^2}$ around $s=0$ to show $P(T_1 = 2n+1) = \dfrac{1}{2n+1}\dbinom{2n}{n}\dfrac{1}{4^n}$.

(i) Use Stirling's approximation to show $E[T_1] = \infty$.

Chapter 3

Markov Chains

3.1 Definition and Construction of the Path Space

Definition (Transition probability). Let $(S, \mathcal{S})$ be a measurable space. We say $p : S \times \mathcal{S} \to [0,1]$ is a **transition probability** on $(S, \mathcal{S})$ if:

(1) For each $x \in S$, $A \mapsto p(x, A)$ is a probability measure on $(S, \mathcal{S})$.

(2) For each $A \in \mathcal{S}$, $x \mapsto p(x, A)$ is a $(S, \mathcal{S})$-measurable function.

Definition (Markov chain). Let $(\mathcal{F}_n)_{n \in \mathbb{Z}_{\geq 0}}$ be a filtration on $(\Omega, \mathcal{F}, P)$. An $(\mathcal{F}_n)$-adapted S-valued process $X = (X_n)_{n \in \mathbb{Z}_{\geq 0}}$ is a **time-homogeneous Markov chain** with respect to $(\mathcal{F}_n)$ and transition probability p if for all $n \in \mathbb{Z}_{\geq 0}$ and $A \in \mathcal{S}$:

$$P(X_{n+1} \in A | \mathcal{F}_n)(\omega) = p(X_n(\omega), A) \quad \text{a.s.} \tag{MP I}$$

The probability distribution μ of X_0 is the **initial distribution**. An equivalent formulation (using FMCT) is: for all $f \in b\mathcal{S}$,

$$E[f(X_{n+1}) | \mathcal{F}_n] = \int_S f(y)\, p(X_n, \mathrm{d}y) \quad \text{a.s.}$$

Exercise. Show that if X is a Markov chain with respect to $(\mathcal{F}_n)$, it is also a Markov chain with respect to its natural filtration $(\mathcal{F}_n^X)$.

We now address the *existence* of Markov chains. We construct the path space and the probability measure on it.

Definition (Product σ-field). Let $(\Omega_i, \mathcal{F}_i)_{i \in I}$ be measurable spaces. Let $\prod_{i \in I} \Omega_i$ be the cartesian product and $\pi_i : \prod_{j \in I} \Omega_j \to \Omega_i$ the projection. A **measurable rectangle** is a set $\bigcap_{i \in F} \pi_i^{-1}(A_i)$ for some finite $F \subset I$ and $A_i \in \mathcal{F}_i$. The **product σ-field** $\bigotimes_{i \in I} \mathcal{F}_i$ is the smallest σ-field containing all measurable rectangles; equivalently, it is the smallest σ-field making all projections π_i measurable.

Lemma (Uniqueness of measures on product spaces). Let $\mathcal{P}$ be a π-system on Ω, and P, Q probability measures on $(\Omega, \sigma(\mathcal{P}))$ with $P(A) = Q(A)$ for all $A \in \mathcal{P}$. Then $P = Q$ on $\sigma(\mathcal{P})$.

Proof. $\mathcal{L} = \{A \in \sigma(\mathcal{P}) : P(A) = Q(A)\}$ is a λ-system containing $\mathcal{P}$; apply the π-λ theorem. □

Theorem (Product measure construction). Let $(\Omega_1, \mathcal{F}_1, \mu_1)$ be a probability space and $(\Omega_2, \mathcal{F}_2)$ a measurable space. Let $\mu_2 : \Omega_1 \times \mathcal{F}_2 \to [0, 1]$ be such that:

(a) $A \mapsto \mu_2(\omega_1, A)$ is a probability measure on $(\Omega_2, \mathcal{F}_2)$ for all ω_1.

(b) $\omega_1 \mapsto \mu_2(\omega_1, A)$ is $(\Omega_1, \mathcal{F}_1)$-measurable for all $A \in \mathcal{F}_2$.

Then there exists a unique probability measure μ on $(\Omega_1 \times \Omega_2, \mathcal{F}_1 \otimes \mathcal{F}_2)$ such that

$$\mu(A \times B) = \int_A \mu_2(\omega_1, B)\, \mu_1(\mathrm{d}\omega_1), \quad A \in \mathcal{F}_1,\ B \in \mathcal{F}_2.$$

Explicitly: $\mu(F) = \displaystyle\int_{\Omega_1} \mu_2(\omega_1, F(\omega_1))\, \mu_1(\mathrm{d}\omega_1)$ where $F(\omega_1) = \{\omega_2 : (\omega_1, \omega_2) \in F\}$.

Proof. **Uniqueness:** follows from Lemma 3.1 applied to the π-system of measurable rectangles.
Existence: One first shows (using the π-λ theorem) that:

- $F(\omega_1) \in \mathcal{F}_2$ for all ω_1 and all $F \in \mathcal{F}_1 \otimes \mathcal{F}_2$ (by checking this for rectangles and extending via λ-system argument).
- $\omega_1 \mapsto \mu_2(\omega_1, F(\omega_1))$ is $\mathcal{F}_1$-measurable for all F (again by λ-system argument on rectangles first).

Then $\mu(F) := \displaystyle\int_{\Omega_1} \mu_2(\omega_1, F(\omega_1))\, \mu_1(\mathrm{d}\omega_1)$ is well-defined and countably additive (by MCT for disjoint unions), hence a probability measure. □

Theorem (Fubini for product measures). Under the hypotheses of Theorem 3.1, let $(\Omega, \mathcal{F}) = (\Omega_1 \times \Omega_2, \mathcal{F}_1 \otimes \mathcal{F}_2)$.

(a) For all $f \in m\mathcal{F}$ and $\omega_1 \in \Omega_1$: $\omega_2 \mapsto f(\omega_1, \omega_2)$ is $\mathcal{F}_2$-measurable.

(b) For all $f \in b\mathcal{F}$: $\omega_1 \mapsto \displaystyle\int_{\Omega_2} f(\omega_1, \omega_2)\, \mu_2(\omega_1, \mathrm{d}\omega_2)$ is bounded $\mathcal{F}_1$-measurable.

(c) For all $f \in b\mathcal{F}$:

$$\int_\Omega f\, \mathrm{d}\mu = \int_{\Omega_1} \left(\int_{\Omega_2} f(\omega_1, \omega_2)\, \mu_2(\omega_1, \mathrm{d}\omega_2) \right) \mu_1(\mathrm{d}\omega_1).$$

Proof. (a) The set of F for which (a) holds is a λ-system containing all rectangles.
(b) Let $\mathcal{H}_1 = \{f \in b\mathcal{F} : \omega_1 \mapsto \int f \,\mathrm{d}\mu_2(\omega_1, \cdot)$ is bounded $\mathcal{F}_1$-measurable$\}$. By (a) the integrals are finite. For rectangles $\mathbf{1}_{A_1 \times A_2}$: $\int \mathbf{1}_{A_2} \,\mathrm{d}\mu_2(\omega_1, \cdot) = \mu_2(\omega_1, A_2)$ is $\mathcal{F}_1$-measurable by hypothesis; multiply by $\mathbf{1}_{A_1}(\omega_1)$. $\mathcal{H}_1$ is a vector space closed under bounded monotone limits (by MCT), so by FMCT, $\mathcal{H}_1 = b\mathcal{F}$.
(c) Similarly define $\mathcal{H}_2 = \{f \in b\mathcal{F} : (c) \text{ holds}\}$; it satisfies the FMCT hypotheses. □

These theorems extend by induction to n-fold products (Theorem 3.6 in notes), and then to countably infinite products.

Theorem (Infinite product construction (Ionescu-Tulcea)). Let $(\Omega_j, \mathcal{F}_j)$, $j \in \mathbb{N}$, be measurable spaces. Let μ_1 be a probability measure on $(\Omega_1, \mathcal{F}_1)$, and for each $j \in \mathbb{N}$ and $(\omega_1, \ldots, \omega_j) \in \prod_{i=1}^{j} \Omega_i$, let $A \mapsto \mu_{j+1}(\omega_1, \ldots, \omega_j, A)$ be a probability measure on $(\Omega_{j+1}, \mathcal{F}_{j+1})$, measurable in $(\omega_1, \ldots, \omega_j)$. Then there exists a unique probability measure P on $\left(\prod_{j=1}^{\infty} \Omega_j, \bigotimes_{j=1}^{\infty} \mathcal{F}_j\right)$ such that for all $n \in \mathbb{N}$ and $B \in \bigotimes_{j=1}^{n} \mathcal{F}_j$:

$$P(\{\omega : (\omega_1, \ldots, \omega_n) \in B\}) = \int_{\Omega_1} \mu_1(\mathrm{d}\omega_1) \int_{\Omega_2} \mu_2(\omega_1, \mathrm{d}\omega_2) \cdots \int_{\Omega_n} \mathbf{1}_B(\omega_1, \ldots, \omega_n) \, \mu_n(\omega_1, \ldots, \omega_{n-1}, \mathrm{d}\omega_n).$$

Proof. The algebra $\mathcal{C}$ of measurable cylinders (sets determined by finitely many coordinates) generates $\bigotimes_j \mathcal{F}_j$. The finite-dimensional consistency (well-definedness on $\mathcal{C}$) follows from the semigroup property of the transition kernels. P is finitely additive on $\mathcal{C}$.
By Carathéodory's extension theorem (and Lemma 3.1 below), it suffices to show: if measurable cylinders $C_n \downarrow \emptyset$, then $P(C_n) \downarrow 0$. Assume for contradiction $P(C_n) \downarrow \delta > 0$. We use a diagonal argument: for each n, extract compact approximations of the base of C_n, build a consistent sequence of points $(\omega_1^*, \omega_2^*, \ldots)$ belonging to all C_n, contradicting $\bigcap C_n = \emptyset$. □

Lemma. Let $\mathcal{A}$ be an algebra on Ω and μ a finitely additive set function on $\mathcal{A}$. If $C_n \downarrow \emptyset$ ($C_n \in \mathcal{A}$) implies $\mu(C_n) \downarrow 0$, then μ is countably additive on $\mathcal{A}$.

Proof. Let (A_n) be pairwise disjoint with $A := \bigsqcup A_n \in \mathcal{A}$. Let $B_n = A \setminus \bigsqcup_{i=1}^{n} A_i \in \mathcal{A}$. Then $B_n \downarrow \emptyset$, so $\mu(B_n) \downarrow 0$, i.e. $\mu(A) - \sum_{i=1}^{n} \mu(A_i) \to 0$. □

Theorem (Existence and construction of Markov chains). Let $(S, \mathcal{S})$ be a measurable space, p a transition probability on $(S, \mathcal{S})$, and μ_0 a probability measure on $(S, \mathcal{S})$. Let $(\Omega, \mathcal{F}) = \left(\prod_{j=0}^{\infty} S, \bigotimes_{j=0}^{\infty} \mathcal{S}\right)$, and $X_n : \Omega \to S$ the projection $X_n(\omega) = \omega_n$. Let $\mathcal{F}_n = \sigma(X_0, X_1, \ldots, X_n)$.

Then there exists a probability measure P on $(\Omega, \mathcal{F})$ such that (X_n) is an $(\mathcal{F}_n)$-Markov chain with transition probability p and initial distribution μ_0.

Proof. Apply Theorem 3.1 with $(\Omega_j, \mathcal{F}_j) = (S, \mathcal{S})$, $\mu_1 = \mu_0$, and $\mu_{n+1}(\omega_0, \ldots, \omega_{n-1}, A) = p(\omega_{n-1}, A)$. The resulting measure P satisfies: for any measurable rectangle $A_0 \times \cdots \times A_n \times S \times S \times \cdots$,

$$P(X_0 \in A_0, \ldots, X_n \in A_n) = \int_{A_0} \mu_0(\mathrm{d}\omega_0) \int_{A_1} p(\omega_0, \mathrm{d}\omega_1) \cdots \int_{A_n} p(\omega_{n-1}, \mathrm{d}\omega_n).$$

The Markov property (MP I) is verified using the π-λ theorem: for $A \in \mathcal{S}$, both $\omega \mapsto P(X_{n+1} \in A | \mathcal{F}_n)(\omega)$ and $\omega \mapsto p(X_n(\omega), A)$ agree on all measurable rectangles in $\mathcal{F}_n$ (by the product formula above), hence agree a.s. by Lemma 3.1. □

3.2 Extensions of the Markov Property

Notation. Given transition probability p and initial distribution μ, let P_μ denote the probability measure on the path space $(\Omega, \mathcal{F}) = (S^{\mathbb{Z}_{\geq 0}}, \mathcal{S}^{\otimes \mathbb{Z}_{\geq 0}})$ from Theorem 3.1. Write $P_x = P_{\delta_x}$, $E_x = E_{\delta_x}$.

Proposition. For all $\varphi \in b\mathcal{S}^{\otimes \mathbb{Z}_{\geq 0}}$, the map $x \mapsto E_x[\varphi]$ is $\mathcal{S}$-measurable, and $E_\mu[\varphi] = \displaystyle\int_S E_x[\varphi]\, \mu(\mathrm{d}x)$.

Proof. By FMCT: the class of φ for which $x \mapsto E_x[\varphi]$ is measurable is a vector space closed under bounded monotone limits; it contains all indicators of measurable rectangles (by the product formula). Hence it equals $b\mathcal{S}^{\otimes \mathbb{Z}_{\geq 0}}$. The integral formula follows similarly. □

Definition (Shift operators). For $n \in \mathbb{Z}_{\geq 0}$, define $\theta_n : \Omega \to \Omega$ by $\theta_n(\omega) = (\omega_n, \omega_{n+1}, \ldots)$.

Exercise. Verify $\theta_n : (\Omega, \mathcal{F}) \to (\Omega, \mathcal{F})$ is measurable.

The following formula is a direct consequence of the product measure construction: for $f_0, f_1, \ldots, f_n \in b\mathcal{S}$ and $x_0 \in S$,

$$E_{x_0}\left[\prod_{i=0}^{n} f_i(X_i)\right] = f_0(x_0) \int_S p(x_1, \mathrm{d}x_2) \cdots \int_S \prod_{i=1}^{n} f_i(x_i)\, p(x_{n-1}, \mathrm{d}x_n). \tag{3.1}$$

Theorem (Markov property (MP II)). Let $\Phi \in b\mathcal{F}$ and μ an initial distribution. Then

$$E_\mu[\Phi \circ \theta_n | \mathcal{F}_n](\omega) = E_{X_n(\omega)}[\Phi] \quad P_\mu\text{-a.s.}$$

Proof. Since $x \mapsto E_x[\Phi]$ is $\mathcal{S}$-measurable (Proposition 3.2), $\omega \mapsto E_{X_n(\omega)}[\Phi]$ is $\mathcal{F}_n$-measurable. It suffices to verify $\displaystyle\int_A (\Phi \circ \theta_n)\, \mathrm{d}P_\mu = \int_A E_{X_n}[\Phi]\, \mathrm{d}P_\mu$ for all $A \in \mathcal{F}_n$.
By the π-λ theorem, it suffices to check this for A a measurable rectangle in $\mathcal{F}_n$ and $\Phi = \mathbf{1}_B$ for B a measurable rectangle. Both sides then reduce to the same expression via the product formula (3.1). The general case follows by FMCT. □

Remark. By setting $n = 1$ and $\Phi(\omega) = \mathbf{1}_B(\omega_1)$ for $B \in \mathcal{S}$ in Theorem 3.2, we recover (MP I) from Definition 3.1.

Theorem (Strong Markov property (SMP)). Let μ be an initial distribution and $(T_n)_{n\in\mathbb{Z}_{\geq 0}}$ a uniformly bounded stochastic process ($\exists\, M > 0$: $|\Phi_n(\omega)| \leq M$ for all n, ω). Let T be an $(\mathcal{F}_n)$-stopping time. Then

$$E_\mu\big[\Phi_T \circ \theta_T\, \mathbf{1}_{\{T<\infty\}} \,\big|\, \mathcal{F}_T\big] = E_{X_T(\omega)}[\Phi_T]\, \mathbf{1}_{\{T<\infty\}} \quad P_\mu\text{-a.s.}$$

Proof. Let $A \in \mathcal{F}_T$. Decompose $\{T < \infty\} = \bigsqcup_{n=0}^{\infty} \{T = n\}$:

$$\int_A \Phi_T \circ \theta_T\, \mathbf{1}_{\{T<\infty\}}\, \mathrm{d}P_\mu = \sum_{n=0}^{\infty} \int_{A\cap\{T=n\}} \Phi_n \circ \theta_n\, \mathrm{d}P_\mu.$$

Since $A \cap \{T = n\} \in \mathcal{F}_n$, by Theorem 3.2:

$$\int_{A\cap\{T=n\}} \Phi_n \circ \theta_n\, \mathrm{d}P_\mu = \int_{A\cap\{T=n\}} E_{X_n}[\Phi_n]\, \mathrm{d}P_\mu.$$

Summing: $\displaystyle\int_A \Phi_T \circ \theta_T\, \mathbf{1}_{\{T<\infty\}}\, \mathrm{d}P_\mu = \int_A E_{X_T}[\Phi_T]\, \mathbf{1}_{\{T<\infty\}}\, \mathrm{d}P_\mu$, proving the SMP. □

3.3 Countable State Markov Chains

For the remainder of Section 3, assume S is countable and $\mathcal{S} = 2^S$.

Definition (Transition matrix). $p : S \times S \to [0, 1]$ is a **transition (stochastic) matrix** if $\displaystyle\sum_{j\in S} p(i, j) = 1$ for all $i \in S$. It defines a transition probability $\displaystyle p(i, A) = \sum_{j\in A} p(i, j)$.

For each $i \in S$, there exists a unique P_i on the path space $(S^{\mathbb{Z}_{\geq 0}}, \mathcal{S}^{\otimes\mathbb{Z}_{\geq 0}})$ such that (X_n) is a Markov chain with transition p and $X_0 = i$ a.s.

Proposition (Chapman–Kolmogorov equation). Define $p^n(i, j) := P_i(X_n = j)$, so $p^1(i, j) = p(i, j)$. For all $m, n \in \mathbb{N}$, $i, j \in S$:

$$p^{m+n}(i, j) = \sum_{k\in S} p^m(i, k)\, p^n(k, j).$$

Proof.

$$\begin{aligned}
p^{m+n}(i, j) = P_i(X_{m+n} = j) &= \sum_{k\in S} P_i(X_{m+n} = j,\, X_m = k) \\
&= \sum_{k\in S} E_i\big[\mathbf{1}_{\{X_m=k\}} E_i[\mathbf{1}_{\{X_{m+n}=j\}} | \mathcal{F}_m]\big] \quad \text{(tower)} \\
&= \sum_{k\in S} E_i\big[\mathbf{1}_{\{X_m=k\}} E_{X_m}[\mathbf{1}_{\{X_n=j\}}]\big] \quad \text{(MP II)} \\
&= \sum_{k\in S} E_i\big[\mathbf{1}_{\{X_m=k\}} p^n(k, j)\big] = \sum_{k\in S} p^m(i, k)\, p^n(k, j).
\end{aligned}$$

□

Note: p^n is the n-th matrix power of p.

Example. (1) **Birth-death chain:** $S = \mathbb{Z}$, $p(i,j) = 0$ for $|i-j| > 1$. Let $p_i = p(i, i+1)$, $q_i = p(i, i-1)$, $r_i = p(i,i) = 1 - p_i - q_i$.

(2) **Branching process:** $S = \mathbb{Z}_{\geq 0}$, $(Y_{n,k})$ i.i.d., $X_0 = x_0$, $X_{n+1} = \sum_{k=1}^{X_n} Y_{n,k}$ (empty sum $= 0$). Transition matrix: $p(i,j) = P(\sum_{k=1}^{i} Y_{0,k} = j)$.

Notation. $T_j = \min\{n > 0 : X_n = j\}$ (first hitting time), $T_j^k = k$-th hitting time. $T_j^0 = 0$. Each T_j^k is a stopping time.

Definition (Recurrence and transience). Let $\rho_{ij} = P_i(T_j < \infty)$. State i is **recurrent** if $\rho_{ii} = 1$; **transient** if $\rho_{ii} < 1$.

Proposition. For all $i, j \in S$, $k \in \mathbb{N}$: $P_i(T_j^k < \infty) = \rho_{ij}\rho_{jj}^{k-1}$.

Proof. Induction on k. The base case $k = 1$ is the definition. For $k \geq 2$: on $\{T_j^{k-1} < \infty\}$, $T_j^k = T_j^{k-1} + T_j \circ \theta_{T_j^{k-1}}$. By the SMP at T_j^{k-1} and $X_{T_j^{k-1}} = j$:

$$P_i(T_j^k < \infty) = E_i\left[\mathbf{1}_{\{T_j^{k-1} < \infty\}} P_{X_{T_j^{k-1}}}(T_j < \infty)\right] = P_j(T_j < \infty)\, P_i(T_j^{k-1} < \infty) = \rho_{jj}\, \rho_{ij}\rho_{jj}^{k-2}.$$

□

Definition. $N(j) = \sum_{n=0}^{\infty} \mathbf{1}_{\{X_n = j\}}$ = number of visits to j.

Corollary. Let $j \in S$.

(i) If j is recurrent, then $N(j) = \infty$ P_j-a.s.

(ii) If j is transient, then under P_j, $N(j) \sim \mathrm{Geom}(1 - \rho_{jj})$. In particular, $E_j[N(j)] = \dfrac{1}{1 - \rho_{jj}}$.

(iii) j is transient $\Leftrightarrow N(j) < \infty$ P_j-a.s. $\Leftrightarrow \sum_{n=0}^{\infty} p^n(j,j) < \infty$.

Proof. (i) By Proposition 3.3: $P_j(N(j) > k) = P_j(T_j^k < \infty) = \rho_{jj}^k = 1^k = 1$ for all k.
(ii) $P_j(N(j) = k) = P_j(N(j) > k-1) - P_j(N(j) > k) = \rho_{jj}^{k-1}(1 - \rho_{jj})$.
(iii) The first equivalence follows from (i),(ii). For the second: by Fubini, $E_j[N(j)] = \sum_{n=0}^{\infty} P_j(X_n = j) = \sum_{n=0}^{\infty} p^n(j,j)$. Combine with (ii). □

Communication. Write $i \to j$ if $\exists\, n \geq 0$: $P_i(X_n = j) > 0$ (equivalently $\rho_{ij} > 0$ or $i = j$). Write $i \leftrightarrow j$ if $i \to j$ and $j \to i$. The equivalence classes under $\leftrightarrow$ are called **communicating classes**.

Exercise. Show $\leftrightarrow$ is an equivalence relation. (Transitivity uses Chapman–Kolmogorov.)

Theorem. Let $i \in S$ be recurrent and $i \to j$. Then j is recurrent and $i \leftrightarrow j$.

Proof. Since i is recurrent: $P_i(X_k = i \text{ i.o.}) = 1$, so $P_i(X_{n+k} = i \text{ i.o.}) = 1$ for any n. Choose $n \geq 0$ with $P_i(X_n = j) > 0$. Then:

$$0 < P_i(X_n = j) = P_i(X_n = j,\, X_{n+k} = i \text{ i.o.}) = P_i(X_n = j) \cdot P_j(X_k = i \text{ i.o.})$$

(using the Markov property at time n and $\{X_n = j, X_{n+k} = i \text{ i.o.}\}$). Hence $P_j(X_k = i \text{ i.o.}) = 1$, so $j \to i$, i.e. $i \leftrightarrow j$.
Now show j is recurrent. Choose m, n with $p^m(j,i) > 0$ and $p^n(i,j) > 0$. By Chapman–Kolmogorov:

$$p^{m+k+n}(j,j) \geq p^m(j,i)\, p^k(i,i)\, p^n(i,j).$$

Summing over k: $\displaystyle\sum_k p^k(j,j) \geq p^m(j,i)\, p^n(i,j) \sum_k p^k(i,i) = \infty$ (since i is recurrent). By Corollary 3.3(iii), j is recurrent. □

Corollary. Recurrence and transience are class properties: if $i \leftrightarrow j$, then i is recurrent iff j is recurrent.

Definition (Closed and irreducible sets). $C \subseteq S$ is **closed** if $P_i(X_n \in C) = 1$ for all $i \in C$, $n \in \mathbb{Z}_{\geq 0}$. Equivalently: $i \in C$, $i \to j$ implies $j \in C$.
C is **irreducible** if $i \leftrightarrow j$ for all $i, j \in C$ (equivalently, $C = [i]$ for all $i \in C$). The Markov chain is **irreducible** if S is irreducible.

Exercise. Show: C is closed $\Leftrightarrow i \not\to j$ for all $i \in C$, $j \notin C \Leftrightarrow p|_{C \times C}$ is a stochastic matrix.

Theorem. (a) Every finite closed set contains a recurrent state.

(b) If S is finite, then S contains a recurrent state.

Proof. (a) Let C be finite and closed. For $i \in C$, $P_i(X_n \in C \text{ for all } n) = 1$, so P_i-a.s. $\displaystyle\sum_{j \in C} N(j) = \infty$. Hence $\displaystyle\sum_{j \in C} E_i[N(j)] = \infty$. Since C is finite, $\exists\, j \in C$ with $E_i[N(j)] = \infty$.
By the SMP:

$$E_i[N(j)] = E_i[\mathbf{1}_{\{T_j < \infty\}} N(j) \circ \theta_{T_j}] = P_i(T_j < \infty)\, E_j[N(j)] = \rho_{ij}\, E_j[N(j)].$$

(Here we use that if $T_j < \infty$ then $X_{T_j} = j$, and apply SMP.) Since $E_i[N(j)] = \infty$ and $\rho_{ij} \leq 1$, we must have $E_j[N(j)] = \infty$. By Corollary 3.3(iii), j is recurrent.
(b) Take $C = S$. □

Definition (Stationary measure and distribution). For a countable Markov chain with transition matrix p:

- A measure μ on S is **stationary** (or **invariant**) for p if $\mu(j) = \sum_{i \in S} \mu(i)p(i,j)$ for all j (i.e. $\mu p = \mu$ when p is viewed as a matrix and μ as a row vector).

- If μ is a probability measure, call it a **stationary distribution**.

- μ is **reversible** for p if $\mu(i)p(i,j) = \mu(j)p(j,i)$ for all $i, j \in S$. Reversibility implies stationarity.

Proposition. If μ is a stationary measure for p, then the Markov chain (X_n) under P_μ is a stationary process: $P_\mu((X_1, X_2, \ldots) \in B) = P_\mu((X_0, X_1, \ldots) \in B)$ for all $B \in \mathcal{S}^{\otimes \mathbb{Z}_{\geq 0}}$. In particular, $P_\mu(X_n \in A) = \mu(A)$ for all $A \in \mathcal{S}$.

Proof. By taking $B = \{X_0 = j\}$ one recovers the definition of stationarity; the general statement follows by the same argument used to prove MP II, replacing $E_\mu[(X_1, X_2, \ldots) \in B]$ and using the product formula. □

Example (Simple random walk on a graph). Let $G = (V, E)$ be a graph without loops or multiple edges, adjacency matrix (a_{ij}) with $a_{ij} \in \{0, 1\}$, $a_{ij} = a_{ji}$, $a_{ii} = 0$. Assume $0 < \mu(i) := \sum_j a_{ij} < \infty$ for all i (degree). Define $p(i,j) = a_{ij}/\mu(i)$. Then μ is reversible: $\mu(i)p(i,j) = a_{ij} = a_{ji} = \mu(j)p(j,i)$.
For the SSRW on $\mathbb{Z}^d$: $\mu(j) = 2d$ for all j, so the counting measure ($\nu(j) = 1$) is stationary.

Theorem (Existence of stationary measure). Let $i \in S$ be recurrent, $T_i = \inf\{n \geq 1 : X_n = i\}$. Define

$$\mu_i(j) = E_i\left[\sum_{n=0}^{T_i - 1} \mathbf{1}_{\{X_n = j\}}\right], \quad j \in S.$$

Then $\mu_i(j) < \infty$ for all j, μ_i is a stationary measure, $\mu_i(i) = 1$, and $\mu_i(j) > 0 \Leftrightarrow i \leftrightarrow j$.

Remark. By Fubini: $\sum_{j \in S} \mu_i(j) = E_i[T_i]$. So $\mu_i / E_i[T_i]$ is a stationary distribution iff $E_i[T_i] < \infty$.

Proof. $\mu_i(i) = 1$ is immediate.
Finiteness and formula. By Fubini:

$$\mu_i(j) = \sum_{n=0}^{\infty} P_i(X_n = j,\, n < T_i). \tag{3.2}$$

Stationarity. Compute $\sum_{j\in S}\mu_i(j)p(j,k)$:

$$\begin{aligned}\sum_{j\in S}\mu_i(j)p(j,k) &= \sum_{n=0}^{\infty}\sum_{j\in S}P_i(X_n=j,\, n<T_i)p(j,k)\\ &= \sum_{n=0}^{\infty}E_i\big[\mathbf{1}_{\{n<T_i\}}E_{X_n}[\mathbf{1}_{\{X_1=k\}}]\big] \quad \text{(tower + MP II)}\\ &= \sum_{n=0}^{\infty}E_i\big[\mathbf{1}_{\{n<T_i\}}\mathbf{1}_{\{X_{n+1}=k\}}\big] = \sum_{n=0}^{\infty}P_i(X_{n+1}=k,\, n<T_i).\end{aligned}$$

Setting $m=n+1$: this equals $\sum_{m=1}^{\infty}P_i(X_m=k,\, m\le T_i) = E_i\left[\sum_{m=1}^{T_i}\mathbf{1}_{\{X_m=k\}}\right] = \mu_i(k)$, using $X_0 = X_{T_i} = i$ P_i-a.s. and recurrence ($T_i<\infty$ a.s.).
$\mu_i(j)>0 \Leftrightarrow i\leftrightarrow j$**.** By (3.2), $\mu_i(j)=0$ iff $P_i(X_n=j, n<T_i)=0$ for all n. If $\mu_i(j)=0$, one can show by SMP that $P_i(T_j>T_i^k)=1$ for all k, hence $P_i(T_j=\infty)=1$, i.e. $j\notin[i]$. Conversely if $j\in[i]$: since i is recurrent, one uses stationarity of μ_i and $p^n(j,i)>0$ to bound $\mu_i(j)$ from below.
Finiteness ($\mu_i(j)<\infty$ for all j). For $j\to i$: choose n with $p^n(j,i)>0$. By stationarity iterated n times: $p^n(j,i)\mu_i(j)\le\mu_i(i)=1$, so $\mu_i(j)\le 1/p^n(j,i)<\infty$. For $j\not\to i$: $\mu_i(j)=0<\infty$. □

Theorem (Uniqueness of stationary measure). Assume $R=[i]$ is a recurrent class. If ν is a stationary measure with $\nu(R^c)=0$, then $\nu=c\mu_i$ for some $c\in[0,\infty)$. Hence μ_i is the unique (up to scalar multiple) stationary measure supported on R.

Proof. We show $\nu(y_0)\ge\nu(i)\mu_i(y_0)$ for all $y_0\in R$. Since ν is a stationary measure, by iterating stationarity n times:

$$\nu(y_0)\ge\nu(i)\sum_{m=1}^{n}P_i(X_1\ne i,\dots,X_{m-1}\ne i, X_m=y_0).$$

Let $n\to\infty$: since i is recurrent, the sum converges to $E_i\left[\sum_{m=1}^{T_i}\mathbf{1}_{\{X_m=y_0\}}\right]=\mu_i(y_0)$. So $\nu(y_0)\ge\nu(i)\mu_i(y_0)$.
Now fix $y_0\in R$ and choose n with $p^n(y_0,i)>0$. By iterating stationarity and using the inequality just proved:

$$\nu(i)=\sum_{y_1\in R}\nu(y_1)p^n(y_1,i)\ge\sum_{y_1\in R}\nu(i)\mu_i(y_1)p^n(y_1,i)=\nu(i)\mu_i(i)=\nu(i).$$

Equality holds throughout, forcing $\nu(y_0)=\nu(i)\mu_i(y_0)$ for all $y_0\in R$. □

Proposition. Let π be a stationary distribution for p and $T=\{i\in S: i \text{ transient}\}$. Then $\pi(T)=0$.

Proof. Suppose $\pi(j) > 0$. We show j is recurrent. Let $S_j = \inf\{n \geq 0 : X_n = j\}$. By stationarity, $\sum_n p^n(i,j)$ is the same for π-a.e. i, and:

$$\infty = \sum_{n=0}^{\infty} \pi(j) = \sum_{i \in S} \pi(i) \sum_{n=0}^{\infty} p^n(i,j) = E_\pi[N(j)] = E_\pi[\mathbf{1}_{\{S_j < \infty\}} N(j) \circ \theta_{S_j}] \leq E_j[N(j)].$$

(using SMP and $X_{S_j} = j$ on $\{S_j < \infty\}$). Hence $E_j[N(j)] = \infty$, so j is recurrent. □

Definition. A recurrent state i is **positive recurrent** if $E_i[T_i] < \infty$; **null recurrent** if $E_i[T_i] = \infty$. These are class properties.

Proposition. Let π be a stationary distribution supported on a recurrent class $R = [j]$. Then for all $j \in R$:

$$\pi(j) = \frac{1}{E_j[T_j]} > 0,$$

and j is positive recurrent.

Proof. Since R is recurrent and $\pi(R) = 1$, by the uniqueness theorem (Theorem 3.3): $\pi = c_j \mu_j$ for some $c_j \geq 0$. Then $1 = \pi(S) = c_j \mu_j(S) = c_j E_j[T_j]$, so $c_j = 1/E_j[T_j]$ and $\pi(j) = c_j \mu_j(j) = 1/E_j[T_j]$. In particular $E_j[T_j] < \infty$. □

Corollary. Positive recurrence is a class property.

Theorem. For a recurrent class R: a stationary distribution supported on R exists if and only if R is positive recurrent (and then it is unique: $\pi(\cdot) = \mu_j(\cdot)/E_j[T_j]$ for any $j \in R$).

Proof. ($\Rightarrow$) Proposition 3.3. ($\Leftarrow$) If $E_j[T_j] < \infty$, then $\pi := \mu_j / E_j[T_j]$ is a stationary distribution by Theorem 3.3. □

Example (Continued from Example 3.3). If G is connected and infinite, S is irreducible. Any stationary measure equals $c\mu$ for some c (Theorem 3.3); since $\mu(S) = \infty$, there is no stationary distribution, and if recurrent the walk is null recurrent. If G is connected and finite, $\pi(j) = \mu(j)/\mu(S)$ is the unique stationary distribution, and the walk is positive recurrent.

Exercise (Classification of all stationary distributions). Let $S = T \cup R = T \cup NR \cup PR$. Let π be a stationary distribution.

(a) Show a stationary distribution exists iff $PR \neq \emptyset$.

(b) Show $\pi(PR) = 1$.

(c) Write $PR = \bigsqcup_{l \in L} R_l$. Let π_l be the unique s.d. on R_l. Show $\pi = \sum_{l \in L} p_l \pi_l$ where $p_l = \pi(R_l)$, $\sum_l p_l = 1$.

(d) Conversely, any $\pi = \sum_{l \in L} p_l \pi_l$ with $p_l \geq 0$, $\sum p_l = 1$ is a s.d.

Theorem (Convergence theorem 1: ergodic theorem for Markov chains). For all $i, j \in S$, P_i-a.s.:

$$\lim_{n\to\infty} \frac{1}{n} \sum_{m=1}^{n} \mathbf{1}_{\{X_m=j\}} = \frac{1}{E_j[T_j]} \mathbf{1}_{\{T_j<\infty\}}.$$

Proof. **Case 1: j transient.** Then $E_j[T_j] = \infty$. By the SMP argument of the proof of Theorem 3.3(a), $E_i[N(j)] \leq E_j[N(j)] < \infty$ (since j transient). As $N_n(j)/n \leq N(j)/n \to 0$ a.s., the result holds.
Case 2: j recurrent, $i = j$. Let T_j^k be the k-th return time. By the SMP (as in the proof of the SLLN), the inter-return times $T_j^k - T_j^{k-1}$ are i.i.d. with mean $E_j[T_j]$. Let $N_n(j)$ be the number of visits up to time n. Since $T_j^{N_n(j)} \leq n < T_j^{N_n(j)+1}$:

$$\frac{T_j^{N_n(j)}}{N_n(j)} \leq \frac{n}{N_n(j)} \leq \frac{T_j^{N_n(j)+1}}{N_n(j)}.$$

By the SLLN (Corollary 2.2) and the recurrence $N_n(j) \to \infty$ a.s.: both sides $\to E_j[T_j]$, so $n/N_n(j) \to E_j[T_j]$, i.e. $N_n(j)/n \to 1/E_j[T_j]$.
Case 3: j recurrent, $i \neq j$. On $\{T_j = \infty\}$: $N_n(j) = 0$ so ratio $= 0 = \dfrac{1}{E_j[T_j]} \mathbf{1}_{\{T_j=\infty\}}$ (both zero). On $\{T_j < \infty\}$: shifting by the finite time T_j does not affect the asymptotic fraction (Case 2 applies after time T_j). A careful use of the SMP and Case 2 yields the result. □

Corollary. For all $i, j \in S$:

$$\lim_{n\to\infty} \frac{1}{n} \sum_{m=1}^{n} p^m(i,j) = \frac{\rho_{ij}}{E_j[T_j]}.$$

Proof. Since $N_n(j)/n \leq 1$ for all n, by DCT:

$$\lim_{n\to\infty} \frac{1}{n} \sum_{m=1}^{n} p^m(i,j) = \lim_{n\to\infty} E_i\left[\frac{N_n(j)}{n}\right] = E_i\left[\frac{1}{E_j[T_j]} \mathbf{1}_{\{T_j<\infty\}}\right] = \frac{\rho_{ij}}{E_j[T_j]}.$$ □

Theorem (Convergence theorem 2). If j is transient or null recurrent ($E_j[T_j] = \infty$), then $p^n(i,j) \to 0$ for all $i \in S$.

Proof. **Transient case.** By the SMP: $E_i[N(j)] \leq E_j[N(j)] = \dfrac{1}{1-\rho_{jj}} < \infty$. Hence $\displaystyle\sum_{n=0}^{\infty} p^n(i,j) = E_i[N(j)] < \infty$, so $p^n(i,j) \to 0$.
Null recurrent case. This is assigned as homework (see also Durrett). □

Definition (Period). For $j \in S$, let $I_j = \{n \geq 0 : p^n(j,j) > 0\}$. The **period** of j is $d_j = \gcd(I_j)$. j is **aperiodic** if $d_j = 1$. Period is a class property.

Lemma. If $d_j = 1$, there exists $n_0 \geq 1$ such that $p^n(j,j) > 0$ for all $n \geq n_0$.

Proof. See Durrett, Lemma 5.6.5. □

Definition (Total variation distance). For $\mu, \nu \in \mathcal{M}_1(S)$:

$$\|\mu - \nu\|_V = \sup_{A \subseteq S} |\mu(A) - \nu(A)|, \qquad \|\mu - \nu\|_1 = \sum_{j \in S} |\mu(j) - \nu(j)|.$$

One can show $\|\mu - \nu\|_V = \frac{1}{2}\|\mu - \nu\|_1$.

Coupling construction. Define the **product transition matrix** $\overline{p}$ on S^2 by $\overline{p}((x_1, y_1), (x_2, y_2)) = p(x_1, x_2)p(y_1, y_2)$. If $(X_n), (Y_n)$ are independent Markov chains with transition p and initial distributions π_1, π_2, then $Z_n = (X_n, Y_n)$ is a Markov chain with transition $\overline{p}$ and initial distribution $\pi_1 \otimes \pi_2$.
Let $P^{\pi_1 \otimes \pi_2}$ denote the law of (Z_n). Define the **coupling time**

$$T_C = \inf\{n \geq 0 : X_n = Y_n\}.$$

Define the **coupled chain**:

$$\widetilde{Z}_n = (X_n, \widetilde{Y}_n), \quad \widetilde{Y}_n = \begin{cases} Y_n & n \leq T_C, \\ X_n & n > T_C. \end{cases}$$

Lemma. The process $(\widetilde{Y}_n)_{n \geq 0}$ defined above is an $(\mathcal{F}_n^Z)$-Markov chain with transition p and initial distribution π_2.

Proof. $\widetilde{Y}_0 = Y_0 \sim \pi_2$. For the Markov property, compute $P^{\pi_1 \otimes \pi_2}(\widetilde{Y}_{n+1} \in A | \mathcal{F}_n^Z)$ by splitting on $\{n < T_C\}$ and $\{T_C \leq n\}$:

$$\begin{aligned} &= \mathbf{1}_{\{n < T_C\}} P^{\pi_1 \otimes \pi_2}(Y_{n+1} \in A | \mathcal{F}_n^Z) + \mathbf{1}_{\{T_C \leq n\}} P^{\pi_1 \otimes \pi_2}(X_{n+1} \in A | \mathcal{F}_n^Z) \\ &= \mathbf{1}_{\{n < T_C\}} p(Y_n, A) + \mathbf{1}_{\{T_C \leq n\}} p(X_n, A) = p(\widetilde{Y}_n, A). \end{aligned}$$ □

Lemma. If p is irreducible and aperiodic, then $\overline{p}$ is also irreducible (and aperiodic).

Proof. Let $(x_1, y_1), (x_2, y_2) \in S^2$. Choose K, L with $p^K(x_1, x_2) > 0$ and $p^L(y_1, y_2) > 0$. By Lemma 3.3, for M large: $p^{L+M}(x_2, x_2) > 0$ and $p^{K+M}(y_2, y_2) > 0$. Then by Chapman–Kolmogorov:

$$\overline{p}^{K+L+M}((x_1, y_1), (x_2, y_2)) \geq p^K(x_1, x_2) p^{L+M}(x_2, x_2) p^L(y_1, y_2) p^{K+M}(y_2, y_2) > 0.$$ □

Theorem (Coupling theorem). Assume p is irreducible, aperiodic, and recurrent. Let $\pi_1, \pi_2 \in \mathcal{M}_1(S)$.

(a) $P^{\pi_1 \otimes \pi_2}(T_C < \infty) = 1$.

(b) $\|P_{\pi_1}(X_n \in \cdot) - P_{\pi_2}(Y_n \in \cdot)\|_V \leq P^{\pi_1 \otimes \pi_2}(T_C > n) \xrightarrow{n \to \infty} 0$.

Proof. (a) By Lemma 3.3, $\overline{p}$ is irreducible. Fix $x_0 \in S$; since $\overline{p}$ is irreducible and p is recurrent (implying $\overline{p}$ is recurrent), by Theorem 3.3: for any $(x, y) \in S^2$, $P_{(x,y)}(Z_n = (x_0, x_0) \text{ i.o.}) = 1$. By Proposition 3.2: $P^{\pi_1 \otimes \pi_2}(T_{(x_0,x_0)} < \infty) = 1$. Since $T_C \leq T_{(x_0,x_0)}$, (a) follows.
(b) For any $A \subseteq S$, using $\widetilde{Y}_n = X_n$ on $\{T_C \leq n\}$ and Lemma 3.3:

$$\begin{aligned} P_{\pi_1}(X_n \in A) - P_{\pi_2}(\widetilde{Y}_n \in A) &= P^{\pi_1 \otimes \pi_2}(X_n \in A, T_C > n) - P^{\pi_1 \otimes \pi_2}(\widetilde{Y}_n \in A, T_C > n) \\ &\leq P^{\pi_1 \otimes \pi_2}(T_C > n). \end{aligned}$$

By symmetry, $|P_{\pi_1}(X_n \in A) - P_{\pi_2}(Y_n \in A)| \leq P^{\pi_1 \otimes \pi_2}(T_C > n)$. By (a), this $\to 0$. □

Theorem (Convergence theorem 3). Let R be a positive recurrent, aperiodic class with unique stationary distribution π on R. Then for all $i \in R$:

$$\lim_{n \to \infty} \|P_i(X_n \in \cdot) - \pi\|_V = 0.$$

Proof. R is closed (Theorem 3.3). Restrict to $S = R$. By Lemma 3.3, $\overline{p}$ is irreducible. Since $\pi \otimes \pi$ is a stationary distribution for $\overline{p}$ (check), by Theorem 3.3, $\overline{p}$ is positive recurrent. Apply Theorem 3.3 with $\pi_1 = \delta_i$ and $\pi_2 = \pi$; since $P_\pi(X_n \in \cdot) = \pi(\cdot)$:

$$\|P_i(X_n \in \cdot) - \pi\|_V \leq P^{\delta_i \otimes \pi}(T_C > n) \to 0.$$ □

Corollary. If $j \in S$ is aperiodic and positive recurrent, then for all $i \in S$:

$$\lim_{n \to \infty} P_i(X_n = j) = \frac{\rho_{ij}}{E_j[T_j]}.$$

Proof.

$$P_i(X_n = j) = E_i[\mathbf{1}_{\{T_j \leq n\}} p^{n - T_j}(j, j)] \xrightarrow{n \to \infty} P_i(T_j < \infty) \cdot \frac{1}{E_j[T_j]} = \frac{\rho_{ij}}{E_j[T_j]},$$

by DCT and Theorem 3.3 (since $p^{n-T_j}(j, j) \to 1/E_j[T_j]$ by Theorem 3.3). □

Chapter 4

Brownian Motion

4.1 Definition and Construction of Brownian Motion

A random variable X is $N(\mu, \sigma^2)$ (**normal with mean μ and variance σ^2**) if it has density $\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$.

Definition (Brownian motion). An $\mathbb{R}$-valued stochastic process $(B_t)_{t\geq 0}$ on $(\Omega, \mathcal{F}, P)$ is a **standard (1-dimensional) Brownian motion** if:

(1) $B_0 = 0$ a.s.

(2) **Independent, stationary increments:** For all $0 \leq t_0 < t_1 < \cdots < t_N$, the increments $\{B_{t_i} - B_{t_{i-1}} : i = 1, \ldots, N\}$ are independent, and $B_{t_i} - B_{t_{i-1}} \sim N(0, t_i - t_{i-1})$.

(3) **Continuous sample paths:** $t \mapsto B_t(\omega)$ is continuous a.s.

Define the **heat kernel**: $p_t(x) = \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{x^2}{2t}\right)$ = density of $N(0, t)$.

Exercise (Semigroup property). For $s, t > 0$ and $x \in \mathbb{R}$:

$$(p_s * p_t)(x) := \int_{\mathbb{R}} p_s(x-y) p_t(y) \, \mathrm{d}y = p_{s+t}(x). \tag{4.1}$$

(Hint: this is equivalent to: if $Z_s \sim N(0, s)$ and $Z_t \sim N(0, t)$ are independent, then $Z_s + Z_t \sim N(0, s+t)$.)

Lemma (FDD characterisation). Properties (1)+(2) in Definition 4.1 hold if and only if the following (FDD) property holds: for all $0 = t_0 < t_1 < \cdots < t_N$ and $A \in \mathcal{B}(\mathbb{R}^{N+1})$,

$$P((B_{t_0}, B_{t_1}, \ldots, B_{t_N}) \in A) = \int_{\mathbb{R}} \cdots \int_{\mathbb{R}} \mathbf{1}_A(0, y_1, \ldots, y_N) \prod_{i=1}^{N} p_{t_i - t_{i-1}}(y_i - y_{i-1}) \, \mathrm{d}y_1 \cdots \mathrm{d}y_N, \tag{FDD}$$

where $y_0 = 0$.

Proof. (1)+(2) $\Rightarrow$ (FDD): Define $T : \mathbb{R}^{N+1} \to \mathbb{R}^{N+1}$ by $T(x_0, \dots, x_N) = (x_0, x_0+x_1, \dots, x_0+\cdots + x_N)$. Then $(B_{t_0}, \dots, B_{t_N}) = T(B_{t_0}, B_{t_1} - B_{t_0}, \dots, B_{t_N} - B_{t_{N-1}})$, and the substitution $x_i = y_i - y_{i-1}$ yields (FDD).
(FDD) $\Rightarrow$ (1) + (2): Use $T^{-1}(y_0, \dots, y_N) = (y_0, y_1 - y_0, \dots, y_N - y_{N-1})$. Applying (FDD) to sets of the form $A_0 \times \cdots \times A_N$ (after the change of variables $x_i = y_i - y_{i-1}$) shows $B_0 = 0$ a.s. and the increments are independent with the correct distributions. □

Construction of Brownian motion. We define the path space $\Omega = \mathbb{R}^{[0,\infty)} = \{\omega : \omega : [0, \infty) \to \mathbb{R}\}$. For a finite subset $U = \{t_0 < t_1 < \cdots < t_N\} \subset [0, \infty)$ containing 0, let $\pi_U : \Omega \to \mathbb{R}^U$ be the projection $\pi_U(\omega) = \omega|_U$. The **algebra of measurable cylinders** is

$$\mathcal{F}_0 = \{B \subseteq \Omega : \exists\, U \in \mathcal{F}_{\text{fin}},\ A \in \mathcal{B}(\mathbb{R}^U) \text{ such that } B = \pi_U^{-1}(A)\},$$

and $\mathcal{F} = \sigma(\mathcal{F}_0) = \sigma(\{X_t : t \geq 0\})$ where $X_t(\omega) = \omega(t)$.

Theorem (Existence of finite-dimensional distributions). There exists a unique probability measure P on $(\Omega, \mathcal{F})$ satisfying (FDD). In particular, the process $(X_t)_{t\geq 0}$ satisfies properties (1) and (2) of Definition 4.1 under P.

Proof. **Well-definedness:** If $\pi_U^{-1}(A) = \pi_{U'}^{-1}(A')$ for $U \subset U'$, we need $P_U(A) = P_{U'}(A')$. By adding one time point at a time and using the semigroup property (4.1), one verifies this consistency.
Finite additivity on $\mathcal{F}_0$: If $\pi_U^{-1}(A) \cap \pi_U^{-1}(A') = \emptyset$ (i.e. $A \cap A' = \emptyset$), then $P(\pi_U^{-1}(A \cup A')) = P_U(A \cup A') = P_U(A) + P_U(A')$.
Countable additivity (via Carathéodory): By Lemma 3.1, it suffices to show: if measurable cylinders $B_n = \pi_{U_n}^{-1}(A_n) \downarrow \emptyset$, then $P(B_n) \downarrow 0$.
Assume for contradiction $P(B_n) \downarrow \delta > 0$. For each n, choose compact $K_n \subset A_n$ with $P_{U_n}(A_n \setminus K_n) < \delta/2^{n+1}$. Let $\widetilde{K}_n = \{\omega : \omega|_{U_m} \in K_m \text{ for all } m \leq n\}$. These are compact in the product topology and $P(\widetilde{K}_n) \geq \delta/2$. By compactness and a diagonalisation, $\bigcap_n \pi_{U_n}^{-1}(K_n) \neq \emptyset$, contradicting $\bigcap_n B_n = \emptyset$.
Uniqueness follows from Lemma 3.1 applied to the π-system $\mathcal{F}_0$. □

We note an important subtlety: the σ-field $\mathcal{F}$ is too coarse to contain the set of continuous paths.

Lemma. (a) For any $B \in \mathcal{F}$, there exists a countable $T \subset [0, \infty)$ and $A \in \mathcal{B}(\mathbb{R}^T)$ such that $B = \{\omega : \omega|_T \in A\}$.

(b) $\mathcal{C} := \{\omega : t \mapsto \omega(t) \text{ is continuous}\} \notin \mathcal{F}$.

Proof. (a) The collection of sets satisfying (a) is a σ-field containing $\mathcal{F}_0$; hence it contains $\mathcal{F}$.
(b) If $\mathcal{C} \in \mathcal{F}$, by (a) there exists countable T and $A \in \mathcal{B}(\mathbb{R}^T)$ with $\mathcal{C} = \{\omega : \omega|_T \in A\}$. Pick $\tilde{t} \in [0, \infty) \setminus T$ and any $\omega_0 \in \mathcal{C}$. Then the path $\tilde{\omega}(t) = \omega_0(t) + \mathbf{1}_{\{\tilde{t}\}}(t)$ satisfies $\tilde{\omega}|_T = \omega_0|_T \in A$, so $\tilde{\omega} \in \mathcal{C}$—but $t \mapsto \tilde{\omega}(t)$ is not continuous at $\tilde{t}$, a contradiction. □

Definition (Version). Processes $(X_t)_{t\geq 0}$ and $(Y_t)_{t\geq 0}$ on $(\Omega, \mathcal{F}, P)$ are **versions** (or **modifications**) of each other if $P(X_t = Y_t) = 1$ for each $t \geq 0$ (the null set may depend on t). Versions have the same finite-dimensional distributions, so if Y is a continuous version of a process satisfying (FDD), then Y is a Brownian motion.

Theorem (Kolmogorov continuity theorem). Suppose $(X_t)_{t\geq 0}$ satisfies: for some $\alpha, \beta, K > 0$,

$$E[|X_t - X_s|^\beta] \leq K|t-s|^{1+\alpha}, \quad s,t \in [0,\infty),\ |s-t| \leq 1. \tag{M}$$

Then there exists a version $(Y_t)_{t\geq 0}$ of (X_t) such that $t \mapsto Y_t(\omega)$ is continuous for all ω. Moreover, for any $0 < \gamma < \alpha/\beta$, there exists $C_\gamma > 0$ such that for all $T \in \mathbb{N}$ and $\omega \in \Omega$, there exists $\delta(T,\omega) > 0$ with:

$$s,t \in [0,T],\ |t-s| < \delta(T,\omega) \Rightarrow |Y_t(\omega) - Y_s(\omega)| \leq C_\gamma |t-s|^\gamma. \tag{HC$_\gamma$}$$

In other words, $t \mapsto Y_t(\omega)$ is locally Hölder continuous with exponent γ.

Proof. Fix $0 < \gamma < \alpha/\beta$. Let $\mathbb{Q}_2 = \{j/2^n : j,n \in \mathbb{Z}_{\geq 0}\}$ be the non-negative dyadic rationals (dense in $[0,\infty)$).
Step 1: Control oscillations at dyadic scales. By Markov's inequality and (M):

$$\begin{aligned} P\left(\max_{0\leq j<n2^n} \left|X\left(\tfrac{j+1}{2^n}\right) - X\left(\tfrac{j}{2^n}\right)\right| > 2^{-n\gamma}\right) &\leq \sum_{j=0}^{n2^n-1} 2^{n\beta\gamma} E\left[\left|X\left(\tfrac{j+1}{2^n}\right) - X\left(\tfrac{j}{2^n}\right)\right|^\beta\right] \\ &\leq n2^n \cdot 2^{n\beta\gamma} \cdot K \cdot 2^{-n(1+\alpha)} = n \cdot 2^{-n(\alpha-\beta\gamma)}. \end{aligned}$$

Since $\gamma < \alpha/\beta$, $\alpha - \beta\gamma > 0$, so $\sum_n n \cdot 2^{-n(\alpha-\beta\gamma)} < \infty$. By Borel–Cantelli, there exists a null set N_0 such that for all $\omega \notin N_0$, there exists $N(\omega) \in \mathbb{N}$ with: for all $n \geq N(\omega)$,

$$\max_{0\leq j<n2^n} \left|X\left(\tfrac{j+1}{2^n}\right) - X\left(\tfrac{j}{2^n}\right)\right| \leq 2^{-n\gamma}. \tag{4.3}$$

Step 2: Hölder continuity on $\mathbb{Q}_2$. Fix $\omega \notin N_0$ and $T \in \mathbb{N}$; set $\delta(T,\omega) = 2^{-(N(\omega)\vee(T+1))}$. Let $s,t \in \mathbb{Q}_2 \cap [0,T]$ with $0 < t-s < \delta$. Choose $n \geq N(\omega)\vee(T+1)$ with $2^{-(n+1)} \leq t-s < 2^{-n}$. Write $s = j2^{-n} + \sum_{k>n} i_k 2^{-k}$ (binary expansion). Define $s_k = j2^{-n} + \sum_{l=n+1}^{k} i_l 2^{-l}$. By (4.3):

$$|X(s) - X(j2^{-n})| \leq \sum_{k>n} 2^{-k\gamma} \leq \frac{2^{-(n+1)\gamma}}{1-2^{-\gamma}} \leq C'_\gamma |t-s|^\gamma.$$

Similarly $|X(t) - X(j'2^{-n})| \leq C'_\gamma |t-s|^\gamma$ for appropriate j'. Since $j' \in \{j, j+1\}$ (from $2^{-(n+1)} \leq t-s < 2^{-n}$): $|X(j2^{-n}) - X(j'2^{-n})| \leq 2^{-n\gamma} \leq 2^\gamma |t-s|^\gamma$. Combining:

$$|X(t) - X(s)| \leq (2C'_\gamma + 2^\gamma)|t-s|^\gamma \tag{4.8}$$

for all $s,t \in \mathbb{Q}_2 \cap [0,T]$ with $|t-s| < \delta(T,\omega)$.

Step 3: Continuous extension. For $\omega \notin N_0$: $s \mapsto X(s,\omega)$ is uniformly continuous on $\mathbb{Q}_2 \cap [0,T]$ for each T. Define

$$Y(t,\omega) = \lim_{\substack{s \downarrow t \\ s \in \mathbb{Q}_2}} X(s,\omega), \quad t \geq 0,\ \omega \notin N_0; \qquad Y(t,\omega) = 0, \quad \omega \in N_0.$$

Then $Y(\cdot,\omega)$ is continuous, satisfies (HC_γ), and for any fixed t, choosing $s_n \downarrow t$ in $\mathbb{Q}_2$:

$$E[|X(s_n) - X(t)|^\beta] \leq K|s_n - t|^{1+\alpha} \to 0,$$

so $X(s_n) \to X(t)$ in L^β and along a subsequence a.s., giving $Y(t) = X(t)$ a.s. □

Corollary. (1) Standard Brownian motion exists.

(2) If B is any Brownian motion and $0 < \gamma < \dfrac{1}{2}$, there exists $C_\gamma > 0$ such that for all $\omega \in \Omega$, $T \in \mathbb{N}$, there exists $\delta(T,\omega) > 0$ such that (HC_γ) holds for a version of B.

Proof. (1) Let (X_t) be the process from Theorem 4.1. Since $X_t - X_s \sim N(0, t-s)$:

$$E[|X_t - X_s|^\beta] = (t-s)^{\beta/2} E[|X_1|^\beta] = C_\beta |t-s|^{\beta/2}.$$

Taking $\beta > 2$, condition (M) holds with $1 + \alpha = \beta/2$, i.e. $\alpha = \beta/2 - 1 > 0$. By Theorem 4.1, there is a continuous version B of X, and since B satisfies (FDD), B is a Brownian motion.
(2) Take $\beta > 2$ large enough that $\gamma < \dfrac{1}{2} - \dfrac{1}{\beta} = \dfrac{\alpha}{\beta}$. □

4.2 Properties of Brownian Motion

Theorem (Nowhere differentiability). Let

$$\Lambda = \{\omega \in \Omega : \exists\, t \geq 0,\ \exists\, \varepsilon, C > 0 \text{ such that } \forall u \in [t, t+\varepsilon],\ |B(u) - B(t)| \leq C|u - t|\}.$$

Then $P(\Lambda) = 0$.

Corollary (Paley–Wiener–Zygmund, 1933). With probability 1, $t \mapsto B(t,\omega)$ is nowhere differentiable.

Proof. If B is differentiable at t_0 with derivative L, then for small $\varepsilon > 0$ and $u \in [t_0, t_0 + \varepsilon]$: $|B(u) - B(t_0)| \leq (|L| + 1)|u - t_0|$. Hence $\omega \in \Lambda$. So $\{B \text{ differentiable at some } t_0\} \subset \Lambda$, and $P(\Lambda) = 0$ completes the proof. □

Proof of Theorem 4.2. For $N \in \mathbb{N}$, define

$$\Lambda_N = \{\omega : \exists\, t \in [0, N) \text{ such that } |B(u) - B(t)| \leq N|u - t| \text{ for all } u \in [t, t + N^{-1}]\}.$$

Then $\Lambda = \bigcup_{N=1}^{\infty} \Lambda_N$. Fix N and let $n \geq 4N$. For $\omega \in \Lambda_N$, choose t as in the definition and $k \in \mathbb{N}$ with $(k-1)n^{-1} \leq t < kn^{-1}$. For $j = 0, 1, 2$:

$$\left| \frac{k+j}{n} - t \right| \leq \frac{4}{n} \leq \frac{1}{N},$$

so by $\omega \in \Lambda_N$:

$$\left|B\left(\tfrac{k+j+1}{n}\right) - B\left(\tfrac{k+j}{n}\right)\right| \le \left|B\left(\tfrac{k+j+1}{n}\right) - B(t)\right| + \left|B\left(\tfrac{k+j}{n}\right) - B(t)\right| \le 2N \cdot \frac{4}{n} = \frac{8N}{n}.$$

Define $\Lambda_{n,N} = \bigcup_{k=1}^{nN} \bigcap_{j=0}^{2} \left\{ |B(\frac{k+j+1}{n}) - B(\frac{k+j}{n})| \le \frac{8N}{n} \right\}$. Then $\Lambda_N \subset \bigcap_{n \ge 4N} \Lambda_{n,N}$.
By independence of increments:

$$\begin{aligned} P(\Lambda_{n,N}) &\le \sum_{k=1}^{nN} \prod_{j=0}^{2} P\left(\left|B\left(\tfrac{k+j+1}{n}\right) - B\left(\tfrac{k+j}{n}\right)\right| \le \frac{8N}{n} \right) \\ &= nN \left(\int_{-8N/\sqrt{n}}^{8N/\sqrt{n}} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \mathrm{d}x \right)^3 \le nN \left(\frac{16N}{\sqrt{n}} \right)^3 = \frac{16^3 N^4}{\sqrt{n}} \to 0 \end{aligned}$$

as $n \to \infty$. Hence $P(\Lambda_N) \le \liminf_{n\to\infty} P(\Lambda_{n,N}) = 0$. □

Remark. Let $H_\alpha(\omega) = \{t \ge 0 : \exists\, \varepsilon, C > 0 \text{ such that } |B(u) - B(t)| \le C|t-u|^\alpha \ \forall u \in [t, t+\varepsilon]\}$.

- Corollary 4.1(2): $H_\alpha(\omega) = [0, \infty)$ a.s. for $0 < \alpha < \frac{1}{2}$.
- Theorem 4.2 (and a modification of its proof): $H_\alpha(\omega) = \emptyset$ a.s. for $\alpha > 1$.
- For $\alpha = \frac{1}{2}$: $H_{1/2}(\omega) = \emptyset$ a.s. (harder result).

4.3 Brownian Motion as a Gaussian Process

Definition (Gaussian random vector and process). An $\mathbb{R}^d$-valued random variable $X = (X_1, \dots, X_d)$ has **mean zero Gaussian distribution** if there exist i.i.d. $N(0,1)$ r.v.'s $Z_1, \dots, Z_k$ and a $k \times d$ matrix $A = (a_{ij})$ such that $X = ZA$ (row vector times matrix). The **covariance matrix** Γ of X is the $d \times d$ matrix with $\Gamma_{ij} = E[X_i X_j] = \mathrm{Cov}(X_i, X_j)$.
Note $\Gamma = A^\top A$, and the distribution of X depends only on Γ:

$$E[e^{i\langle u, X\rangle}] = \exp\left(-\tfrac{1}{2} u \Gamma u^\top\right), \quad u \in \mathbb{R}^d.$$

A process $(X_t)_{t\ge0}$ is a **mean zero Gaussian process** if all its finite-dimensional distributions are mean zero Gaussian. The **covariance function** $C(s,t) = E[X_s X_t] = \mathrm{Cov}(X_s, X_t)$ uniquely determines all finite-dimensional distributions.

Theorem (Gaussian characterisation of Brownian motion). The following are equivalent:

(a) $(B_t)_{t\ge0}$ is a standard Brownian motion.

(b) $(B_t)_{t\ge0}$ is a mean zero Gaussian process with $C(s,t) = s \wedge t$ for all $s, t \ge 0$, and $t \mapsto B_t$ is a.s. continuous.

Proof. (a)⇒(b): Let $0 < t_1 < t_2$. Write $\begin{pmatrix} B_{t_1} \\ B_{t_2} \end{pmatrix} = \begin{pmatrix} \sqrt{t_1} & 0 \\ \sqrt{t_1} & \sqrt{t_2 - t_1} \end{pmatrix} \begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix}$ where $Z_1 = B_{t_1}/\sqrt{t_1}$ and $Z_2 = (B_{t_2} - B_{t_1})/\sqrt{t_2 - t_1}$ are i.i.d. $N(0,1)$ by Definition 4.1. The same argument applies to any finite collection, so B is a mean zero Gaussian process. For $0 \le s \le t$: $C(s,t) = E[B_s B_t] = E[B_s^2 + B_s(B_t - B_s)] = s + E[B_s]E[B_t - B_s] = s = s \wedge t$ (using independence).
(b)⇒(a): A mean zero Gaussian process with $C(s,t) = s \wedge t$ satisfies (FDD) (since the covariance matrix determines the Gaussian distribution). Thus B satisfies (1)+(2) of Definition 4.1, and continuity is given. Hence B is a Brownian motion. □

Corollary (Symmetry properties of BM). Let (B_t) be a standard Brownian motion. The following are also standard Brownian motions:

(1) **Symmetry:** $(-B_t)_{t \ge 0}$.

(2) **Scaling:** $\left(\frac{1}{a} B_{a^2 t}\right)_{t \ge 0}$ for any $a > 0$.

(3) **Time inversion:** $\widetilde{B}_t = t B_{1/t}$ for $t > 0$, $\widetilde{B}_0 = 0$.

Proof. Each is a mean zero Gaussian process with continuous paths (for (3): continuity on $(0, \infty)$ is clear; continuity at 0 uses $\lim_{t \downarrow 0} t B_{1/t} = 0$ a.s., verified by the law of the iterated logarithm or the characterisation below). The covariance function in each case is $s \wedge t$: e.g. for (2): $E[\frac{1}{a} B_{a^2 s} \cdot \frac{1}{a} B_{a^2 t}] = \frac{1}{a^2} E[B_{a^2 s} B_{a^2 t}] = \frac{1}{a^2}(a^2 s \wedge a^2 t) = s \wedge t$. Apply Theorem 4.3. □

4.4 Wiener Measure

Let $C = C([0,\infty), \mathbb{R}) = \{y : [0,\infty) \to \mathbb{R} : y \text{ continuous}\}$. Define

$$d(y_1, y_2) = \sum_{n=0}^{\infty} \frac{d_n(y_1, y_2) \wedge 1}{2^n}, \quad d_n(y_1, y_2) = \sup_{t \in [0,n]} |y_1(t) - y_2(t)|.$$

Remark. (1) d is a complete separable metric on C (a Polish space). Separability: approximate continuous functions on compact sets by polynomials (Weierstrass).

(2) Let $\mathcal{C}$ be the Borel σ-field on (C, d) (smallest σ-field containing all open sets). Then $\mathcal{C} = \sigma(Y_t : t \ge 0)$ where $Y_t(y) = y(t)$.

Proof. Each Y_t is continuous (hence Borel), so $\sigma(Y_t : t \ge 0) \subset \mathcal{C}$. Conversely, $d_n(y_0, \cdot) = \sup_{t \le n, t \in \mathbb{Q}} |Y_t(\cdot) - y_0(t)|$ is $\sigma(Y_t : t \ge 0)$-measurable (countable sup); hence $d(y_0, \cdot)$ is measurable and open balls belong to $\sigma(Y_t : t \ge 0)$. Since (C, d) is separable, every open set is a countable union of open balls, giving $\mathcal{C} \subset \sigma(Y_t : t \ge 0)$. □

(3) The measurable cylinders $\mathcal{C}_0 = \{B \in \mathcal{C} : \exists N, 0 \le t_0 < \cdots < t_N,\ A \in \mathcal{B}(\mathbb{R}^{N+1}),\ B = \{y : (y_{t_0}, \ldots, y_{t_N}) \in A\}\}$ form an algebra with $\mathcal{C} = \sigma(\mathcal{C}_0)$.

Definition (Wiener measure). Let B be a Brownian motion on $(\Omega, \mathcal{F}, P)$, and redefine $B(\cdot, \omega)$ to be continuous for *all* ω (on the null set where it isn't continuous, set $B \equiv 0$). Define the map $\mathbf{B} : \Omega \to C$ by $\mathbf{B}(\omega) = B(\cdot, \omega)$. This map is $(\mathcal{F}, \mathcal{C})$-measurable (check via the π-λ theorem).
The **Wiener measure** is the pushforward $P_0 = P \circ \mathbf{B}^{-1}$ on $(C, \mathcal{C})$:

$$P_0(D) = P(\mathbf{B} \in D), \quad D \in \mathcal{C}.$$

For $x \in \mathbb{R}$, define $P_x(D) = P(x + \mathbf{B} \in D)$ (Brownian motion started at x). We write E_0, E_x for the corresponding expectations.

4.5 Brownian Motion as a Markov Process and a Martingale

For a filtration $(\mathcal{F}_t)_{t \geq 0}$, define the **right-continuous augmentation** $\mathcal{F}_{t+} = \bigcap_{s>t} \mathcal{F}_s$; this is also a filtration, and it is right-continuous: $(\mathcal{F}_{t+})_{s+} = \mathcal{F}_{t+}$.

Definition ($(\mathcal{F}_t)$-Brownian motion). $(B_t)_{t \geq 0}$ is an $(\mathcal{F}_t)$**-Brownian motion** if:

(1) (B_t) is $(\mathcal{F}_t)$-adapted.

(2) For all $t > s \geq 0$, $B_t - B_s$ is independent of $\mathcal{F}_s$.

Exercise. If B is a Brownian motion, it is an $(\mathcal{F}_t^B)$-Brownian motion.

Proposition (Right-continuous filtrations). If B is an $(\mathcal{F}_t)$-Brownian motion, then B is an $(\mathcal{F}_{t+})$-Brownian motion.

Proof. Let $s < t$ and $s_n \downarrow s$, $s_n > s$. We show $B_t - B_s$ is independent of $\mathcal{F}_{s+} = \bigcap_n \mathcal{F}_{s_n}$.
For any bounded continuous $\varphi : \mathbb{R} \to \mathbb{R}$:

$$E[\varphi(B_t - B_s)|\mathcal{F}_{s+}] = \lim_{n \to \infty} E[\varphi(B_t - B_s)|\mathcal{F}_{s_n}] \quad \text{(DCT)}.$$

Since $B_t - B_{s_n}$ is independent of $\mathcal{F}_{s_n}$ and $B_t - B_s = (B_t - B_{s_n}) + (B_{s_n} - B_s)$ with $B_{s_n} - B_s \to 0$ in L^2:

$$E[\varphi(B_t - B_{s_n})|\mathcal{F}_{s_n}] = E[\varphi(B_t - B_{s_n})] \to E[\varphi(B_t - B_s)].$$

Hence $E[\varphi(B_t - B_s)|\mathcal{F}_{s+}] = E[\varphi(B_t - B_s)]$. By the π-λ theorem (as in the proof of independence), $B_t - B_s$ is independent of $\mathcal{F}_{s+}$. □

Proposition (BM is a martingale). If B is an $(\mathcal{F}_t)$-Brownian motion, then B is an $(\mathcal{F}_t)$-martingale.

Proof. For $0 \leq s < t$: since $B_t - B_s$ is independent of $\mathcal{F}_s$ with mean zero:

$$E[B_t|\mathcal{F}_s] = E[B_t - B_s|\mathcal{F}_s] + E[B_s|\mathcal{F}_s] = E[B_t - B_s] + B_s = 0 + B_s = B_s. \qquad \square$$

Notation. E_0 denotes expectation under Wiener measure P_0.

Theorem (Markov property (MP1)). Let B be an $(\mathcal{F}_t)$-Brownian motion. For $s \geq 0$, let $B^{(s)}(t) = B(s+t) - B(s)$, $t \geq 0$. Then $B^{(s)}$ is a Brownian motion independent of $\mathcal{F}_s$:

$$E_0[\varphi(B^{(s)})|\mathcal{F}_s] = E_0[\varphi], \quad \varphi \in bC.$$

Proof. Fix $A \in \mathcal{F}_s$ with $P_0(A) > 0$. Define $Q_A(\Lambda) = P_0(A \cap \Lambda)/P_0(A)$. We show $B^{(s)}$ is a Brownian motion under Q_A. Since $B^{(s)}$ has a.s. continuous paths under P_0, the same holds under $Q_A \ll P_0$. Also $B^{(s)}(0) = 0$ a.s.
For independent increments of $B^{(s)}$ under Q_A: let $0 < t_0 < t_1 < t_2$, $C_1, C_2 \in \mathcal{B}(\mathbb{R})$:

$$\begin{aligned} &Q_A\Big(B^{(s)}_{t_1} - B^{(s)}_{t_0} \in C_1,\, B^{(s)}_{t_2} - B^{(s)}_{t_1} \in C_2\Big) \\ &= \frac{1}{P_0(A)} P_0(A,\, B_{s+t_1} - B_{s+t_0} \in C_1,\, B_{s+t_2} - B_{s+t_1} \in C_2)\,. \end{aligned}$$

Conditioning first on $\mathcal{F}_{s+t_1}$ (tower property): $B_{s+t_2} - B_{s+t_1}$ is independent of $\mathcal{F}_{s+t_1}$ (since B is an $(\mathcal{F}_t)$-BM), so:

$$= \frac{1}{P_0(A)} P_0(A,\, B_{s+t_1} - B_{s+t_0} \in C_1) \cdot P_0(B_{s+t_2} - B_{s+t_1} \in C_2).$$

Conditioning on $\mathcal{F}_s$: $A \in \mathcal{F}_s$ and $B_{s+t_1} - B_{s+t_0}$ is independent of $\mathcal{F}_s$, so:

$$= P_0(B_{s+t_1} - B_{s+t_0} \in C_1) \cdot P_0(B_{s+t_2} - B_{s+t_1} \in C_2) = Q_A(\cdots) \cdot Q_A(\cdots).$$

The general case for N increments is similar. Hence $B^{(s)}$ has independent $N(0, t_i - t_{i-1})$ increments under Q_A, i.e. $B^{(s)}$ is a BM under Q_A.
Since this holds for all $A \in \mathcal{F}_s$ with $P_0(A) > 0$: $E_0[\varphi(B^{(s)})|\mathcal{F}_s] = E_0[\varphi(B^{(s)})] = E_0[\varphi]$. □

Corollary (Markov property (MP2)). For $\psi : C \to \mathbb{R}$ bounded measurable:

$$E_0[\psi(B(s+\cdot))|\mathcal{F}_s] = E_{B(s,\omega)}[\psi].$$

Proof. Define $\Phi(x, y) = \psi(y + x)$ (well-defined, bounded, measurable). Then $\psi(B(s+\cdot)) = \Phi(B(s), B^{(s)})$. Since $B(s)$ is $\mathcal{F}_s$-measurable and $B^{(s)}$ is independent of $\mathcal{F}_s$ by Theorem 4.5: by the pull-out property and the independence, $E_0[\Phi(B(s), B^{(s)})|\mathcal{F}_s] = E_{B(s,\omega)}[\psi]$. □

Corollary (Blumenthal's 0-1 law). For any $A \in \mathcal{F}^B_{0+} := \bigcap_{\varepsilon > 0} \sigma(\{B_s : 0 \leq s \leq \varepsilon\})$, we have $P_0(A) \in \{0, 1\}$.

Proof. By Proposition 4.5, B is an $(\mathcal{F}^B_{t+})$-BM. By MP1, $B^{(0)} = B$ is independent of $\mathcal{F}^B_{0+}$. Since $A \in \mathcal{F}^B_{0+} \subset \sigma(B_s : s \leq 1)$, A is also in $\sigma(B_s : s \geq 0) = \sigma(B^{(0)})$. Hence A is independent of itself: $P_0(A) = P_0(A \cap A) = P_0(A)^2$. □

Example. By Blumenthal's 0-1 law and the fact that $\limsup_{t \downarrow 0} \dfrac{B(t)}{\sqrt{2t \log\log(1/t)}}$ is $\mathcal{F}^B_{0+}$-measurable (hence a.s. constant), combined with Theorem 8.5.1 in Durrett, this constant equals 1.

Stopping times for continuous-time processes.
$T : \Omega \to [0, \infty]$ is an $(\mathcal{F}_t)$-**stopping time** if $\{T \le t\} \in \mathcal{F}_t$ for all $t \ge 0$.

Exercise. Show $\{T = \infty\} \in \sigma(\bigcup_{t \ge 0} \mathcal{F}_t)$. (Hint: $\{T < \infty\} = \bigcup_{n \in \mathbb{N}} \{T < n\}$.)

Lemma. If $(\mathcal{F}_t)$ is right-continuous and $\{T < t\} \in \mathcal{F}_t$ for all $t > 0$, then T is a stopping time.

Proof. $\{T \le t\} = \bigcap_{k \in \mathbb{N}} \{T < t + 1/k\} \in \bigcap_k \mathcal{F}_{t+1/k} = \mathcal{F}_{t+} = \mathcal{F}_t$. □

Assumption 4.22. For the remainder of Section 4, we assume:

(1) B is an $(\mathcal{F}_t)$-Brownian motion.

(2) $(\mathcal{F}_t)$ is right-continuous (e.g. $\mathcal{F}_t = \mathcal{F}^B_{t+}$).

(3) $t \mapsto B(t, \omega)$ is continuous for all ω (redefine on null set).

Example. For an open set $G \subset \mathbb{R}$: $T_G = \inf\{t \ge 0 : B_t \in G\}$ is an $(\mathcal{F}_t)$-stopping time. Indeed, $\{T_G < t\} = \bigcup_{s<t, s \in \mathbb{Q}} \{B_s \in G\} \in \mathcal{F}_t$; apply Lemma 4.5.

Proposition. Let (T_n) be $(\mathcal{F}_t)$-stopping times.

(i) If $T_n \downarrow T$, then T is a stopping time.

(ii) If $T_n \uparrow T$, then T is a stopping time.

In particular, hitting times of closed sets are stopping times.

Proof. (i) $\{T < t\} = \bigcup_n \{T_n < t\} \in \mathcal{F}_t$; apply Lemma 4.5.

(ii) $\{T \le t\} = \bigcap_n \{T_n \le t\} \in \mathcal{F}_t$.

For closed F: the sets $G_n = \{x : d(x, F) < 1/n\}$ are open, so T_{G_n} are stopping times. One shows $T_{G_n} \uparrow T_F$, so T_F is a stopping time. □

Exercise. Show $\limsup_n T_n$ is an $(\mathcal{F}_t)$-stopping time whenever each T_n is.

Definition (Discrete skeleton and $\mathcal{F}_T$). For an $(\mathcal{F}_t)$-stopping time T, the **discrete skeleton** is

$$T_n(\omega) = \begin{cases} k/2^n & \text{if } k/2^n \le T(\omega) < (k+1)/2^n,\ k \in \mathbb{Z}_{\ge 0}, \\ \infty & T(\omega) = \infty. \end{cases}$$

Properties: (i) $T_n \downarrow T$; (ii) $T_n \in \{k/2^n : k \in \mathbb{Z}_{\ge 0}\} \cup \{\infty\}$; (iii) T_n is a stopping time.
The **stopped σ-field** is $\mathcal{F}_T = \{A \in \mathcal{F} : \forall t \ge 0,\ A \cap \{T \le t\} \in \mathcal{F}_t\}$.

Exercise. Let T be an $(\mathcal{F}_t)$-stopping time.

(i) $\mathcal{F}_T$ is a σ-field.

(ii) If $(\mathcal{F}_t)$ is right-continuous: $\mathcal{F}_T = \{A \in \mathcal{F} : \forall t > 0,\ A \cap \{T < t\} \in \mathcal{F}_t\}$.

(iii) T is $\mathcal{F}_T$-measurable.

Proposition. (a) S, T stopping times with $S \leq T$ a.s. implies $\mathcal{F}_S \subset \mathcal{F}_T$.

(b) Right-continuous filtration: if $T_n \downarrow T$, then $\mathcal{F}_T = \bigcap_n \mathcal{F}_{T_n}$.

Proof. (a) Let $A \in \mathcal{F}_S$, $t \geq 0$. Since $S \leq T$: $\{T \leq t\} \subset \{S \leq t\}$, so $A \cap \{T \leq t\} = (A \cap \{S \leq t\}) \cap \{T \leq t\} \in \mathcal{F}_t$.
(b) By (a), $\mathcal{F}_T \subset \mathcal{F}_{T_n}$ for all n, so $\mathcal{F}_T \subset \bigcap_n \mathcal{F}_{T_n}$. For the reverse: if $A \in \bigcap_n \mathcal{F}_{T_n}$, then $A \cap \{T < t\} = \bigcup_n (A \cap \{T_n < t\}) \in \mathcal{F}_t$ (right-continuous filtration). □

Lemma. Under Assumption 4.22, $\mathbf{1}_{\{T<\infty\}} B_T$ is $\mathcal{F}_T$-measurable.

Proof. For the discrete skeleton T_n: $\mathbf{1}_{\{T_n<\infty\}} B_{T_n} = \sum_{k \geq 0} \mathbf{1}_{\{T_n = k/2^n\}} B_{k/2^n}$ is $\mathcal{F}_{T_n}$-measurable. Since B is continuous and $T_n \downarrow T$: $\mathbf{1}_{\{T<\infty\}} B_T = \lim_{n\to\infty} \mathbf{1}_{\{T_n<\infty\}} B_{T_n}$, which is $\bigcap_n \mathcal{F}_{T_n} = \mathcal{F}_T$-measurable. □

4.6 Strong Markov Property and Applications

Theorem (Strong Markov property). Let B be an $(\mathcal{F}_t)$-BM satisfying Assumption 4.22, and T a stopping time. For all bounded $\mathcal{F}/\mathcal{B}(\mathbb{R})$-measurable $\psi : C \to \mathbb{R}$:

$$E_0[\psi(B(T + \cdot))\, \mathbf{1}_{\{T<\infty\}} | \mathcal{F}_T](\omega) = E_{B(T,\omega)}[\psi]\, \mathbf{1}_{\{T<\infty\}}. \tag{SMP}$$

Proof. **Step 1: ψ bounded continuous.** (We first establish (SMP) for bounded continuous ψ.) Let (T_n) be the discrete skeleton of T; each T_n is countably valued. By the Markov property (MP1/MP2 at countable stopping times, proved exactly as in the discrete case using Theorem 4.5):

$$E_0[\psi(B(T_n + \cdot))\, \mathbf{1}_{\{T_n<\infty\}} | \mathcal{F}_{T_n}](\omega) = E_{B(T_n,\omega)}[\psi]\, \mathbf{1}_{\{T_n<\infty\}}.$$

Since $\mathcal{F}_T = \bigcap_n \mathcal{F}_{T_n}$ (Proposition 4.5(b)):

$$E_0[\psi(B(T_n + \cdot))\, \mathbf{1}_{\{T_n<\infty\}} | \mathcal{F}_T] = E_{B(T_n,\omega)}[\psi]\, \mathbf{1}_{\{T_n<\infty\}}.$$

As $n \to \infty$: $T_n \downarrow T$, so $B(T_n, \omega) \to B(T, \omega)$ by continuity of B, and ψ is continuous, hence $\psi(B(T_n + \cdot)) \to \psi(B(T + \cdot))$ (by continuity of B and ψ). By DCT applied to the conditional expectations:

$$E_0[\psi(B(T + \cdot))\, \mathbf{1}_{\{T<\infty\}} | \mathcal{F}_T] = E_{B(T,\omega)}[\psi]\, \mathbf{1}_{\{T<\infty\}}.$$

Step 2: General $\psi \in bC$. Since F closed in C implies $\psi_n(y) = e^{-nd(y,F)} \to \mathbf{1}_F(y)$ (bounded convergence), and $\mathcal{C}$ is generated by closed sets, the class of ψ for which (SMP) holds contains all bounded continuous functions, is a vector space, is closed under bounded monotone limits, and hence equals bC by FMCT. □

Corollary (Extended SMP). Let $\Psi : \mathbb{R} \times C \to \mathbb{R}$ be bounded measurable. Then

$$E_0[\Psi(T, B(T+\cdot))\,\mathbf{1}_{\{T<\infty\}}|\mathcal{F}_T](\omega) = E_{B(T,\omega)}[\Psi(T(\omega),\cdot)]\,\mathbf{1}_{\{T<\infty\}}.$$

Proof. For $\Psi(u,y) = \psi_1(u)\psi_2(y)$: combine Theorem 4.6 with the $\mathcal{F}_T$-measurability of T. Extend to general Ψ by FMCT. □

Theorem (Reflection principle). Let (B_t) be an $(\mathcal{F}_t)$-BM satisfying Assumption 4.22. Let $M_t = \sup_{s\le t} B_s$. For all $a \ge 0$, $t > 0$:

$$P_0(M_t \ge a) = 2P_0(B_t \ge a).$$

Proof. Let $T_a = \inf\{t \ge 0 : B_t = a\}$. Then T_a is a stopping time (hitting time of a closed set). Note:

- $\{M_t \ge a\} = \{T_a \le t\}$ (since B is continuous and starts at 0).
- $\{M_t \ge a, B_t \ge a\} = \{T_a \le t, B_t \ge a\}$.
- $\{M_t \ge a, B_t < a\} = \{T_a < t, B_t < a\}$ (if $M_t \ge a$ and $B_t < a$, then $T_a < t$).

We claim $P_0(M_t \ge a, B_t < a) = P_0(M_t \ge a, B_t \ge a)$. Indeed:

$$\begin{aligned}
P_0(T_a < t, B_t < a) &= E_0[\mathbf{1}_{\{T_a<t\}} P_0(B_t < a|\mathcal{F}_{T_a})] \quad \text{(tower)}\\
&= E_0[\mathbf{1}_{\{T_a<t\}} P_{B(T_a)}(B_{t-T_a} < a)] \quad \text{(SMP)}\\
&= E_0[\mathbf{1}_{\{T_a<t\}} P_a(B_{t-T_a} < a)] \quad (B_{T_a} = a \text{ on } \{T_a < t\})\\
&= E_0[\mathbf{1}_{\{T_a<t\}} P_0(B_{t-T_a} < 0)] \quad \text{(shift by } a)\\
&= \tfrac{1}{2} P_0(T_a < t) \quad \text{(symmetry of BM, } P_0(B_s < 0) = \tfrac{1}{2})
\end{aligned}$$

and similarly $P_0(T_a \le t, B_t \ge a) = \dfrac{1}{2} P_0(T_a \le t)$. Therefore:

$$P_0(M_t \ge a) = P_0(T_a \le t) = 2P_0(T_a \le t, B_t \ge a) \le 2P_0(B_t \ge a).$$

Also $P_0(B_t \ge a) \le P_0(M_t \ge a)$ trivially, and $P_0(B_t = a) = 0$, giving the equality. □

Corollary (Distribution of the hitting time). For $a \ne 0$, the stopping time $T_a = \inf\{t \ge 0 : B_t = a\}$ has density

$$f_a(s) = \frac{|a|}{\sqrt{2\pi}\, s^{3/2}} \exp\left(-\frac{a^2}{2s}\right) \mathbf{1}_{(0,\infty)}(s).$$

In particular, $P_0(T_a < \infty) = 1$ but $E_0[T_a^p] < \infty \Leftrightarrow p \in (0, \dfrac{1}{2})$.

Proof. By symmetry $T_a \overset{d}{=} T_{-a}$, so WLOG $a > 0$. By Theorem 4.6 and $B_t \sim N(0,t)$:

$$P_0(T_a \leq t) = P_0(M_t \geq a) = 2P_0(B_t \geq a) = 2\int_a^\infty \sqrt{\frac{1}{2\pi t}} e^{-x^2/(2t)} \mathrm{d}x.$$

Differentiating with respect to t (or substituting $x = a/\sqrt{s}\cdot\sqrt{t}$): let $u = a/\sqrt{t}$, $\mathrm{d}u = -\dfrac{a}{2t^{3/2}}\mathrm{d}t$:

$$\frac{\mathrm{d}}{\mathrm{d}t} P_0(T_a \leq t) = \frac{a}{\sqrt{2\pi}\, t^{3/2}} e^{-a^2/(2t)} = f_a(t). \qquad \square$$

Exercise. For $p > 0$ and $a > 0$: $E_0[T_a^p] < \infty \Leftrightarrow p < \dfrac{1}{2}$.

Appendix

Lemma. Let $f : [0, \infty) \to \mathbb{R}$ with $f(0) = 0$, f continuous on $(0, \infty)$. If $\lim_{n\to\infty} f(q_n) = 0$ for every non-increasing sequence (q_n) in $\mathbb{Q} \cap (0, \infty)$ with $q_n \downarrow 0$, then f is continuous at 0.

Proof. Suppose f is not continuous at 0. Then $\exists\, \varepsilon > 0$ and $r_n \downarrow 0$ with $|f(r_n)| \geq \varepsilon$. Since f is continuous at $r_n \in (0, \infty)$, choose $q_n \in (r_n/2, 2r_n) \cap \mathbb{Q}$ with $|f(r_n) - f(q_n)| \leq \varepsilon/2$, so $|f(q_n)| \geq \varepsilon/2$. By passing to a subsequence, $q_n \downarrow 0$. By hypothesis $f(q_n) \to 0$, contradicting $|f(q_n)| \geq \varepsilon/2 > 0$. $\square$